

FinCompass User Manual

From First Principles to Probabilistic Market Analysis

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August 2026

Document version: v1.2.0
Applicable software release: FinCompass v1.2.0
Status: Current
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Abstract

This manual is written for two readers at once: a person encountering investing and probability for the first time, and a technically trained reader who wants to inspect the equations and implementation assumptions. It begins with elementary ideas such as price, return, volume, probability, risk and dollar-cost averaging. It then develops the three distinct statistical layers used by FinCompass: a Bayesian evidence score, a frozen empirical probability of forward benchmark outperformance, and a bounded adaptive Bayesian residual that reacts to new information while preserving the validated anchor. Every major equation is followed by a worked numerical example showing where the quantities come from and how the answer is calculated. A separate section introduces the Wyckoff accumulation–markup–distribution–markdown framework as a descriptive market-structure overlay. The manual shows how a lay reader can combine business quality, valuation, uncertainty, forecast calibration, market structure and portfolio risk when deciding whether a security is merely worth watching, suitable for routine dollar-cost averaging, a candidate for staged accumulation, better held, or a candidate for partial trimming. The Wyckoff overlay is not represented as a validated return-prediction model in the current software. Synthetic examples are used for teaching and do not constitute investment recommendations or evidence of future market performance.

Intended audience: first-time investors, students, practitioners, quantitative researchers, instructors, and reviewers.

Educational scope: This document explains a research and decision-support framework. It is not individualized investment advice, does not guarantee profit, and does not protect against loss.

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1 Orientation

FinCompass has enough mathematics to look intimidating if the equations are presented before the questions they answer. This manual therefore uses the reverse order. Each topic begins with a practical question, then introduces the smallest amount of mathematics needed to answer it, then shows the implemented equation, and finally discusses the assumptions that make the equation reasonable or fragile.

A reader can use the manual in three ways. A first-time reader can stay with the paragraphs marked *Beginner checkpoint* and the worked examples. A practitioner can also read the implementation notes, case studies and action checklists. A mathematically trained reader can move directly to Sections 5–14 and the appendices, where the equations are written in their full form.

1.1 Five Separate Questions

Most investing mistakes begin by combining questions that are not the same question. FinCompass deliberately separates at least five of them:

1. **Is this a financially strong company or asset?** This is an evidence question.
2. **Is the current valuation and macro context supportive or demanding?** This is a context question.
3. **What is the calibrated probability of a precisely defined future event?** This is a forecasting question.
4. **What has changed since the frozen forecast was produced?** This is a realtime evidence question.
5. **What should a person do with that information?** This is a decision question and depends on risk tolerance, time horizon, diversification, liquidity needs, position size and personal circumstances in addition to the model output.

The fifth question is the one for which no universal equation exists. Two people can rationally take different actions from the same forecast because one may already own a large position while the other owns none. For that reason the manual describes *decision-support categories* rather than pretending that a probability alone determines “buy” or “sell.”

Beginner checkpoint. A probability is not an instruction. If FinCompass estimates a 65% probability of a defined outperformance event, it is saying something about uncertainty in that event under the model and information contract. It is not saying “put 65% of your money into the stock.”

1.2 Three Distinct Quantities

FinCompass contains three different statistical objects.

Evidence score

A 0–10 Bayesian research summary. The probability statements around this score concern uncertainty in the *score itself*.

Anchor probability

A frozen empirical forecast of a specific forward benchmark-outperformance event. It is allowed to be interpreted as a return-event probability only after its data and locked-test validation meet the required governance tier.

Adaptive probability

A candidate probability obtained by adding a bounded Bayesian residual to the frozen anchor when new information arrives. The residual is applied live only after its own prequential performance, calibration, temporal-breadth, freshness and drift gates pass.

A high evidence score can coexist with a mediocre forecast probability. A good company can be expensive. A weak company can rally. A strong forecast can also be wrong on any individual case. The architecture is designed to preserve those distinctions rather than hide them.

1.3 Wyckoff as an Interpretation Layer

The Wyckoff method describes market behavior using recurring ideas of accumulation, markup, distribution and markdown, along with concepts such as springs, signs of strength, upthrusts and last points of support or supply [Wyckoff, 1910, StockCharts ChartSchool, 2026]. In this manual it is used as a *market-structure vocabulary*. It helps a non-specialist ask whether price and volume behavior is consistent with demand absorbing supply, a trend already under way, supply appearing near a mature advance, or a downtrend still dominating.

The current FinCompass application does **not** contain a validated Wyckoff phase classifier, and the manual does not relabel the Wyckoff framework as one. Section 10 shows how common observations can be measured quantitatively, but those measurements are kept outside the validated return-probability engine unless and until they receive separate point-in-time validation.

1.4 Dollar-Cost Averaging

Dollar-cost averaging (DCA) usually means investing equal or planned amounts at regular intervals, regardless of short-term price movement [U.S. Securities and Exchange Commission, Investor.gov, 2026]. That is different from receiving a lump sum today and intentionally holding some of it in cash to stage entry over several months. Vanguard's historical analysis found that immediate lump-sum investment beat temporary cost averaging about two-thirds of the time in the periods and markets it studied, largely because money held in cash gives up expected risk premium [Wimmer and Zorn, 2023]. Cost averaging can nevertheless reduce short-term timing risk and regret for a loss-averse investor.

This distinction matters throughout the manual. FinCompass can help a user decide whether to *maintain*, *pause*, or *cautiously accelerate* a pre-existing contribution plan, but it does not change the basic fact that DCA is primarily a disciplined contribution process rather than a magical market-timing rule.

Important limitation. No chart pattern, evidence score, Bayesian probability or DCA schedule guarantees a profit. A DCA plan also does not protect against loss in a security whose business permanently deteriorates. The case studies deliberately include examples where averaging down is the wrong response.

2 System Overview

Figure 1 shows the complete learning path used in the rest of the manual.

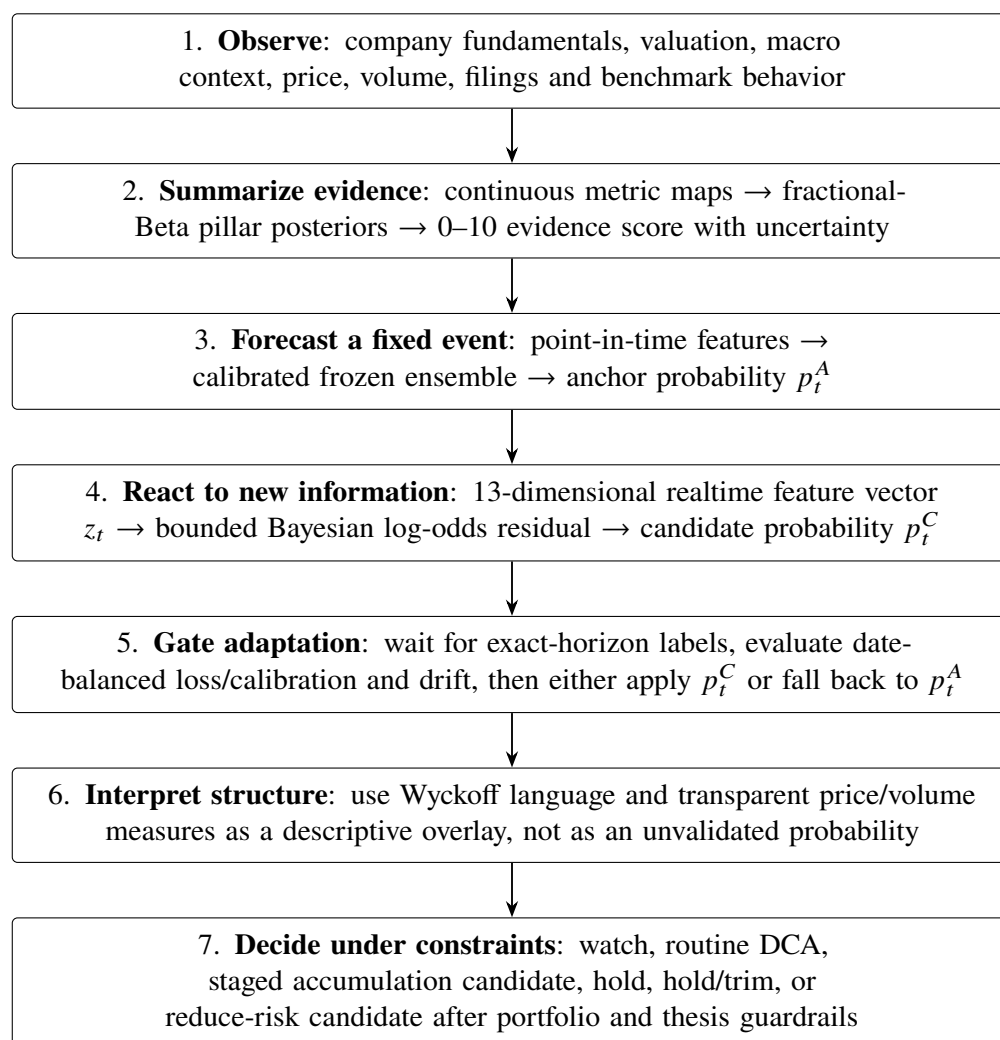


Figure 1: The FinCompass learning and decision-support workflow. The final decision layer is intentionally separated from the probability model because personal constraints are not contained in the market data.

The rest of the manual takes these seven boxes apart, works through the arithmetic, and puts them back together in progressively harder examples.

3 Financial and Mathematical Foundations

This section assumes no prior finance or statistics knowledge. Readers who are already comfortable with returns, probability, odds and logarithms can skim it, but the definitions are used exactly later in the manual.

3.1 Price and Value

A stock price is the amount at which buyers and sellers exchange one share at a particular time. It is not, by itself, a statement about whether the company is cheap, expensive, safe or risky. A share can fall from \$100 to \$50 and still be overvalued if the business has deteriorated more than the price. A share can rise from \$50 to \$100 and still be attractive if earnings, cash flow and competitive position improved even faster.

That is why FinCompass does not use “price went down” as a standalone accumulation signal.

3.2 Return

If a price starts at P_0 and ends at P_1 , the simple return is

$$R = \frac{P_1}{P_0} - 1. \quad (1)$$

Worked example. A stock rises from \$80 to \$92.

$$R = \frac{92}{80} - 1 = 1.15 - 1 = 0.15 = 15\%.$$

A 15% return means the value of the position increased by 15% before taxes, fees and other effects.

If the stock falls from \$80 to \$64,

$$R = \frac{64}{80} - 1 = -0.20 = -20\%.$$

Notice that a 20% loss requires a 25% gain to return to the starting value: $64 \times 1.25 = 80$. This asymmetry is one reason risk management matters.

3.3 Benchmark-relative return

A stock can rise and still underperform its opportunity set. If a stock gains 8% while a benchmark gains 12%, its simple excess return is

$$R^{\text{excess}} = 8\% - 12\% = -4\%. \quad (2)$$

The strict FinCompass forecasting target asks whether a stock outperforms a benchmark over a fixed future horizon. This is more precise than asking whether the stock merely goes up.

3.4 Volume

Volume is the number of shares traded over a period. In Wyckoff analysis, price movement is often read together with volume because a large move on unusually high volume can carry a different interpretation from the same move on very light volume. Volume is not automatically bullish when high or bearish when low. It describes trading effort; the result in price must also be examined.

3.5 Range Position

Suppose the highest price over a chosen window is H , the lowest is L , and the current price is P . A simple range-position measure is

$$r = \frac{P - L}{H - L}. \quad (3)$$

If $r = 0$, the price is at the bottom of the observed range. If $r = 1$, it is at the top.

Worked example. Over the last three months a stock traded between \$40 and \$60. Today it is \$45.

$$r = \frac{45 - 40}{60 - 40} = \frac{5}{20} = 0.25.$$

The price is 25% of the way up from the range low. That fact alone is not a buy signal; it becomes useful only when combined with evidence about demand, supply, business condition and broader risk.

3.6 Mean and Median

The arithmetic mean of values $x_1, \dots, x_n$ is

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i.$$

The median is the middle value after sorting. A single extreme observation can move the mean substantially while leaving the median nearly unchanged. FinCompass therefore uses robust peer medians and interquartile ranges in places where outliers are common.

3.7 Standardization and z-Scores

A z-score tells us how unusual a value is relative to a reference distribution. With mean μ and standard deviation σ ,

$$z = \frac{x - \mu}{\sigma}. \quad (4)$$

A value of $z = 2$ is two standard deviations above the reference mean. In realtime features FinCompass uses standardized recent-return and volume measures so that “unusual” can be compared across securities with different normal trading scales.

3.8 Probability

A probability p lies between 0 and 1. A value of $p = 0.70$ means 70%, but the interpretation is not “the event will happen.” It means the model assigns probability 0.70 to the defined event, given its assumptions and information.

Good probability models are judged over many forecasts. If a well-calibrated model issues many forecasts near 70%, roughly 70% of those events should occur over a suitable evaluation set. That idea is called calibration and is developed in Section 8.

3.9 Odds and log-odds

Probability can be expressed as odds:

$$\text{odds} = \frac{p}{1 - p}. \quad (5)$$

At $p = 0.50$, odds equal 1. At $p = 0.75$, odds equal $0.75/0.25 = 3$, often described as 3-to-1 odds.

The log-odds, or logit, is

$$\text{logit}(p) = \ln\left(\frac{p}{1 - p}\right). \quad (6)$$

The inverse operation is the sigmoid function

$$\text{sigmoid}(x) = \frac{1}{1 + e^{-x}}. \quad (7)$$

Why use log-odds? Because evidence can be added in log-odds space while the final transformed probability automatically stays between 0 and 1.

Worked example. Suppose $p = 0.58$. The odds are

$$\frac{0.58}{0.42} = 1.3810.$$

The log-odds are

$$\ln(1.3810) \approx 0.323.$$

If new validated evidence contributes +0.33 log-odds, the new log-odds are $0.323 + 0.33 = 0.653$. Convert back:

$$p' = \frac{1}{1 + e^{-0.653}} \approx 0.658.$$

So the candidate probability moves from 58% to about 65.8%. Section 7 explains why FinCompass may still refuse to apply that candidate live.

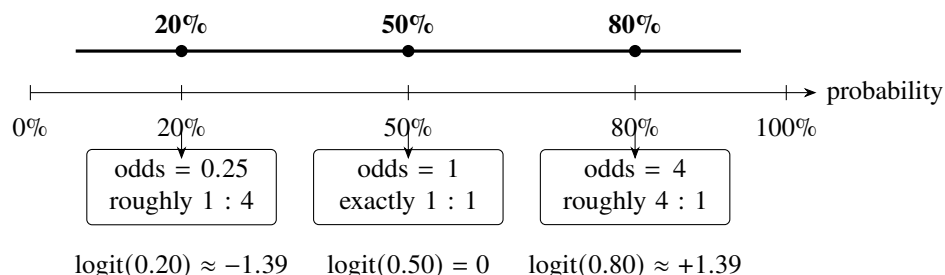


Figure 2: Probability, odds and log-odds describe the same belief on different scales. A beginner can read the top line; the model performs additive updates in log-odds because that scale is unbounded while the converted probability always remains between 0 and 1.

3.10 Uncertainty

A model that says “I am uncertain” is often more useful than a model that always produces a sharp answer. FinCompass reports intervals, evidence coverage, calibration and gate status because the amount of information behind a number matters. A 7.8 score built from nearly all expected data is not equivalent to a 7.8 score built from two surviving metrics.

Beginner checkpoint. The goal is not to eliminate uncertainty. Markets do not permit that. The goal is to measure uncertainty, avoid pretending that missing information is known, and make decisions whose downside remains acceptable when the forecast is wrong.

4 Using FinCompass

The user interface is designed around separate research tasks rather than a single master score. The exact labels may evolve over time, but the logic is organized around research/evidence, screening/comparison, frozen forecasting, realtime adaptation and settings/governance.

4.1 First-Use Workflow

For a first analysis, use the following sequence:

1. Enter or select a ticker.
2. Read the source and freshness information before interpreting the score.
3. Review the five evidence pillars individually instead of jumping to the composite.
4. Check the evidence coverage and 90% model interval.
5. If a live-eligible frozen forecast model is installed, read the anchor probability and its model/version information.
6. Open the Live workspace only after understanding the anchor. Compare anchor and candidate probability, source freshness, top adaptive contributions and gate status.

7. Use Wyckoff structure, if desired, only as a separate chart-reading overlay described later in this manual.
8. Make a portfolio decision only after considering position size, diversification, cash needs, risk tolerance and whether the business thesis remains intact.

Important limitation. The distributed validation fixtures are intentionally marked `fixture_only`. They prove that the statistical and software machinery works on controlled data; they do not authorize a live market probability. A clean FinCompass installation may therefore show no live-eligible market forecast until a real point-in-time model earns the appropriate validation tier.

4.2 Evidence Pillars

FinCompass uses five weighted pillars in the composite evidence score:

Pillar	Weight	Question	Interpretation
Quality	25%	Is the business producing strong economic results?	ROE, ROIC, margins, cash flow and growth-related evidence.
Durability (moat)	20%	Does performance appear economically durable?	Persistence/proxy evidence designed to complement rather than simply duplicate quality.
Safety	20%	Can the balance sheet and cash-generation profile absorb stress?	Leverage, liquidity and coverage evidence.
Valuation	20%	How demanding is the current price relative to available valuation evidence?	Absolute and, where enough peers exist, robust sector-relative multiples.
Cycle	15%	Is the current valuation/macro/commodity regime supportive or demanding?	Current context only; it is explicitly not a recession clock or return forecast.

The weights sum to 1:

$$0.25 + 0.20 + 0.20 + 0.20 + 0.15 = 1.00.$$

The composite is not a simple hand-calculated weighted average of point estimates because FinCompass propagates posterior draws and examines more than one dependence assumption. Section 5 derives that calculation.

4.3 Evidence coverage

Coverage tells you how much of the expected metric weight was actually observed. Suppose a pillar expects total weight 5.0 but only metrics totaling 3.5 are available. Then

$$\text{coverage} = \frac{3.5}{5.0} = 0.70 = 70\%.$$

The missing 30% is not filled with optimistic guesses. The Bayesian prior keeps the posterior closer to neutral and the uncertainty wider.

4.4 Confidence labels

The High/Medium/Low confidence label is a compact summary of both evidence coverage and interval width. It should be read as a statement about *model/evidence precision*, not as confidence that a stock price will rise.

4.5 Forecast workspace

The frozen forecast estimates a precisely defined event. Under the strict reference profile, that event is whether the stock outperforms SPY over the next 252 common trading sessions with a zero excess-return hurdle. Other configurations can change horizon, benchmark and hurdle, but changing the target contract requires rebuilding and retraining rather than silently reusing a model trained for a different question.

A forecast record should therefore always be read together with:

- model ID and validation tier;
- horizon and benchmark;
- as-of date/time and feature contract;
- probability and uncertainty information;
- calibration/performance evidence;
- whether the model is eligible for live use.

4.6 Live workspace

The Live workspace answers: *what changed since the anchor forecast?* It can display fresh market behavior, filings and macro updates, the anchor probability, the adaptive candidate, the bounded log-odds shift, posterior shift uncertainty, top feature contributions, source health and the online gate state.

A candidate probability is not automatically the applied probability. The applied value follows the fail-closed rule developed later:

$$p_t^{\text{live}} = \begin{cases} p_t^C, & \text{if all live-use gates pass,} \\ p_t^A, & \text{otherwise.} \end{cases}$$

4.7 Settings

Settings are deliberately split between convenience options and model-defining options. Changing a display preference should not create a new statistical model. Changing the forecast horizon, benchmark, hurdle, calibration method or adaptive learning contract can change the meaning of the model and therefore creates a new lineage or requires retraining.

This distinction is one of the strongest habits a beginner can learn: *if you change the question, you have changed what counts as a valid answer.*

5 Bayesian Evidence Scoring

The evidence plane answers a research question: *given the information currently available, how strong is the company-and-regime evidence under the FinCompass rubric, and how uncertain is that conclusion?* It does not estimate the probability of future profit or benchmark outperformance.

5.1 Continuous Metric Scoring

Raw financial metrics use different units. ROIC may be a percentage, a P/E ratio may be a multiple, and interest coverage may be another multiple. Before evidence can be combined, each metric is mapped to a common 0–10 scale using continuous piecewise-linear interpolation.

If raw value x lies between knots (a_k, s_k) and (a_{k+1}, s_{k+1}) , the interpolated score is

$$f(x) = s_k + \frac{x - a_k}{a_{k+1} - a_k} (s_{k+1} - s_k). \quad (8)$$

Worked example. For ROIC, two implementation knots are $8\% \mapsto 6.4$ and $12\% \mapsto 8.0$. Suppose a company has ROIC of 10%.

$$\begin{aligned} f(0.10) &= 6.4 + \frac{0.10 - 0.08}{0.12 - 0.08} (8.0 - 6.4) \\ &= 6.4 + \frac{0.02}{0.04} (1.6) \\ &= 6.4 + 0.8 \\ &= 7.2. \end{aligned}$$

The raw 10% ROIC therefore contributes an absolute rubric score of 7.2 rather than jumping abruptly between bins.

This continuity removes threshold cliffs. A company at 11.99% and one at 12.01% should not receive materially different evidence merely because a hand-written boundary was crossed.

5.2 Robust Peer Comparison

When enough suitable sector peers are available, FinCompass supplements the absolute rubric with a robust peer-relative score. For peer median m and interquartile range q , define

$$z^{\text{IQR}} = d \frac{x - m}{\max(q, \varepsilon)}, \quad (9)$$

where $d = 1$ if larger is better and $d = -1$ if smaller is better.

The peer score is

$$s^{\text{peer}} = \text{clip}(5 + 2.2z^{\text{IQR}}, 1, 9.5). \quad (10)$$

With peer weight ρ , the blended score is

$$s = (1 - \rho)s^{\text{abs}} + \rho s^{\text{peer}}. \quad (11)$$

Worked example. Continue the ROIC example. Suppose the sector median ROIC is 8% and the peer IQR is 4 percentage points. The company has 10% ROIC, so

$$z^{\text{IQR}} = \frac{0.10 - 0.08}{0.04} = 0.5.$$

The peer score is

$$5 + 2.2(0.5) = 6.1.$$

The absolute score was 7.2. With the default 25% peer blend,

$$s = 0.75(7.2) + 0.25(6.1) = 5.4 + 1.525 = 6.925.$$

The result is about 6.93. The peer information moderates the absolute rubric without overwhelming it.

If too few peers exist, FinCompass omits the peer term rather than pretending a tiny comparison group is stable.

5.3 Fractional-Beta Evidence Aggregation

Normalize the 0–10 score to $q = s/10$. A score of 7.5 becomes $q = 0.75$. FinCompass then uses a neutral Beta prior

$$U \sim \text{Beta}(\alpha_0, \beta_0), \quad \alpha_0 = \beta_0 = 2.$$

For observed metric j with rubric score q_j and evidence weight w_j , the implementation multiplies the weight by evidence scale $c = 4$ and updates

$$\alpha = \alpha_0 + c \sum_j w_j q_j, \quad (12)$$

$$\beta = \beta_0 + c \sum_j w_j (1 - q_j). \quad (13)$$

The pillar score is the posterior mean mapped back to 0–10:

$$S = 10 \frac{\alpha}{\alpha + \beta}. \quad (14)$$

Worked example. Suppose only three metrics are available in a simplified quality example:

Metric	0–10 score	Evidence weight	$q = s/10$
Metric 1	8.5	1.10	0.85
Metric 2	7.4	1.35	0.74
Metric 3	6.0	1.00	0.60

Start with $\alpha_0 = \beta_0 = 2$. Each weight is multiplied by $c = 4$.

For Metric 1, effective weight is 4.4. It adds

$$4.4(0.85) = 3.74 \quad \text{to } \alpha, \quad 4.4(0.15) = 0.66 \quad \text{to } \beta.$$

For Metric 2, effective weight is 5.4. It adds

$$5.4(0.74) = 3.996 \quad \text{to } \alpha, \quad 5.4(0.26) = 1.404 \quad \text{to } \beta.$$

For Metric 3, effective weight is 4.0. It adds

$$4(0.60) = 2.40 \quad \text{to } \alpha, \quad 4(0.40) = 1.60 \quad \text{to } \beta.$$

Therefore

$$\alpha = 2 + 3.74 + 3.996 + 2.40 = 12.136,$$

$$\beta = 2 + 0.66 + 1.404 + 1.60 = 5.664.$$

The posterior pillar score is

$$S = 10 \frac{12.136}{12.136 + 5.664} \approx 6.82.$$

The point estimate is 6.82, not the raw weighted average, because the neutral prior shrinks sparse evidence toward 5.

For this illustrative posterior $\text{Beta}(12.136, 5.664)$, the central 90% interval mapped to 0–10 is approximately [4.93, 8.46]. The interval is deliberately wide because the example uses only three metrics.

5.4 Evidence Coverage

Coverage is kept separate from the score. If the pillar definition expects weight W^{exp} and observed metrics contribute W^{obs} ,

$$C = \min \left(1, \frac{W^{\text{obs}}}{W^{\text{exp}}} \right). \quad (15)$$

If expected weight is 6.5 and observed weight is 3.45,

$$C = \frac{3.45}{6.5} \approx 0.531 = 53.1\%.$$

A 6.82 score with 53% coverage should be read much more cautiously than the same score with 95% coverage.

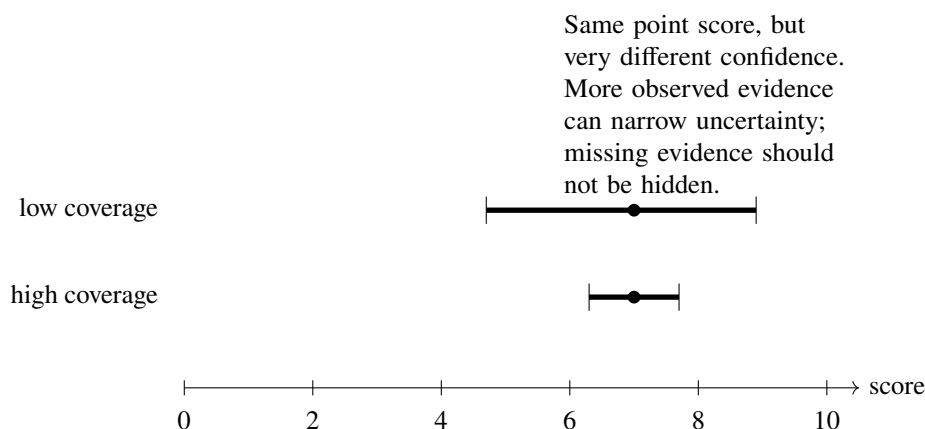


Figure 3: Illustrative evidence-coverage effect. The intervals are teaching examples, not a fixed FinCompass lookup table. The important lesson is that a point score and the amount of evidence supporting it are separate quantities.

5.5 Dependence-Sensitive Composite Uncertainty

FinCompass weights the five pillars as

$$(w_Q, w_D, w_S, w_V, w_C) = (0.25, 0.20, 0.20, 0.20, 0.15). \quad (16)$$

For a simple point-estimate illustration, suppose the pillar means are

$$(7.5, 7.0, 8.0, 5.5, 6.0).$$

The weighted mean would be

$$\begin{aligned} S_{\text{simple}} &= 0.25(7.5) + 0.20(7.0) + 0.20(8.0) + 0.20(5.5) + 0.15(6.0) \\ &= 1.875 + 1.4 + 1.6 + 1.1 + 0.9 \\ &= 6.875. \end{aligned}$$

The actual implementation goes further. It draws from each Beta pillar posterior and evaluates two dependence scenarios:

1. independent pillar draws;
2. perfectly positively rank-coupled (comonotonic) draws.

The reported 90% model interval is the envelope of the two scenarios. This is a sensitivity analysis for unknown positive dependence, not a claim that either dependence structure is the true market-generating process.

Beginner checkpoint. Why do this? Quality, durability and cycle can reuse related business information. If the same underlying good news influences several pillars, treating them as completely independent can make the final interval look more precise than it really is.

5.6 Cycle Pillar vs. Wyckoff Cycle

FinCompass's built-in cycle pillar is a current-regime evidence measure based on valuation regime, a bounded quality buffer, optional yield-curve and high-yield spread context, and optional commodity context. It is not a Wyckoff phase detector and it does not predict a recession date.

The implementation explicitly prevents a favorable macro reading from becoming an independent return forecast. Section 9 introduces Wyckoff as a separate chart-structure layer.

Implementation note. The equations in this section correspond to `services/scoring.py` and constants in `config.py`. In the current implementation, the five pillar weights are 25/20/20/20/15 percent; the Beta prior is (2, 2); the evidence scale is 4; the reported central interval level is 90%; and the Monte Carlo propagation uses 6000 draws. Internal component identifiers are implementation metadata and are deliberately not used as product version numbers in this manual.

6 Forecast Probability

The evidence score and the forecast probability answer different questions. A research score asks how strong the current evidence looks under a rubric. A forecast probability must define a future event, preserve the historical information available at each observation, train a model, calibrate its probabilities, and then show that those probabilities behave acceptably on data that were not used for fitting.

6.1 Target Definition

For stock i , benchmark b , observation time t and horizon H , let $\tau_H(t)$ be the H th future trading session for which both stock and benchmark have usable prices. Define

$$R_{i,t}^{(H)} = \frac{P_i(\tau_H(t))}{P_i(t)} - 1, \quad (17)$$

$$R_{b,t}^{(H)} = \frac{P_b(\tau_H(t))}{P_b(t)} - 1. \quad (18)$$

For excess-return hurdle η ,

$$Y_{i,t} = \mathbb{I} \left\{ R_{i,t}^{(H)} - R_{b,t}^{(H)} > \eta \right\}. \quad (19)$$

The strict reference profile uses $H = 252$, benchmark SPY and $\eta = 0$.

Worked example. A stock is \$100 at the observation date. At the exact 252nd common future session it is \$118. SPY moves from \$500 to \$560 over the same common-session endpoints.

$$R_i = \frac{118}{100} - 1 = 18\%,$$

$$R_b = \frac{560}{500} - 1 = 12\%.$$

Excess return is $18\% - 12\% = 6\%$. With hurdle $\eta = 0$, the label is $Y = 1$ because 6% is greater than 0.

If the stock had ended at \$109, its return would be 9%, excess return would be -3% , and $Y = 0$.

The endpoint is tied to the original forecast date. If a background process runs late, it must still resolve the label at the exact H th common session; it cannot use a later price because that would silently change the target.

6.2 Point-in-Time Features

The reference feature set includes returns over 21/63/126/252 sessions, stock-minus-benchmark relative returns over the same windows, annualized realized volatility, benchmark returns and volatility, 126/252-session drawdowns, distance from the 52-week high, the 50/200-day moving-average relationship and a 63-session standardized volume measure. Optional SEC fundamentals are attached using the date on which the filing information became publicly available.

A feature observed at time t must satisfy a simple rule:

$$\text{available_time}(x) \leq t. \quad (20)$$

This is the mathematical form of “no hindsight.”

6.3 Predictor Standardization

The Bayesian logistic component replaces missing training values by training-set medians, subtracts each feature's training mean, and divides by its training standard deviation. For feature x_j ,

$$\tilde{x}_j = \frac{x_j - \mu_j}{\sigma_j}. \quad (21)$$

Standardization makes a coefficient comparable across features measured in very different units.

6.4 Logistic Probability Model

Let $\tilde{x}$ be the standardized feature vector, with an intercept appended. The logistic model is

$$p = \text{sigmoid}(\beta_0 + \beta_1 \tilde{x}_1 + \cdots + \beta_d \tilde{x}_d). \quad (22)$$

Worked example. Use a deliberately small three-feature teaching model:

$$\beta_0 = 0.10, \quad \beta_1 = 0.45, \quad \beta_2 = 0.30, \quad \beta_3 = -0.25.$$

Suppose standardized values are

$$\tilde{x}_1 = 0.8, \quad \tilde{x}_2 = 0.5, \quad \tilde{x}_3 = 1.2.$$

The linear predictor is

$$\begin{aligned} z &= 0.10 + 0.45(0.8) + 0.30(0.5) - 0.25(1.2) \\ &= 0.10 + 0.36 + 0.15 - 0.30 \\ &= 0.31. \end{aligned}$$

The probability is

$$p = \frac{1}{1 + e^{-0.31}} \approx 0.577.$$

The illustrative model therefore assigns about 57.7% probability to the defined event.

The coefficients above are teaching numbers, not coefficients from a live FinCompass market model.

6.5 Bayesian Prior and MAP Estimation

FinCompass's Bayesian logistic component places zero-centered Gaussian priors on coefficients:

$$\beta_j \sim N(0, \sigma_\beta^2). \quad (23)$$

The strict reference profile uses $\sigma_\beta = 1.5$ for feature coefficients and a wider prior for the intercept.

For labels $y_i \in \{0, 1\}$ and probabilities p_i , the negative log posterior minimized by the MAP estimate is

$$\mathcal{L}(\beta) = \sum_i \left[\log(1 + e^{x_i^\top \beta}) - y_i x_i^\top \beta \right] + \frac{1}{2} \beta^\top \Lambda \beta, \quad (24)$$

where Λ is the prior-precision matrix.

The first term penalizes predictions that disagree with observed labels. The second term penalizes unnecessarily large coefficients. This regularization is particularly important in noisy financial data.

6.6 Laplace Posterior Approximation

After finding the MAP coefficients $\hat{\beta}$, define the executable curvature matrix

$$W = \text{diag}\{\max(p_i(1 - p_i), 10^{-8})\}.$$

The small floor is a numerical safeguard near probability boundaries. The Hessian is

$$H = X^\top W X + \Lambda. \quad (25)$$

The Laplace approximation uses

$$\beta \mid D \approx N(\hat{\beta}, H^{-1}). \quad (26)$$

Drawing coefficient vectors from this approximation and passing them through the logistic function produces posterior probability intervals for the Bayesian component.

6.7 Calibrated Ensemble

The frozen anchor combines three complementary components:

1. Bayesian logistic regression for a regularized, interpretable linear log-odds relation;
2. histogram gradient boosting for nonlinear thresholds/interactions;
3. a regularized random forest for a different form of nonlinear partitioning.

Each component is probability-calibrated on a chronological validation stage. A later validation stage learns nonnegative ensemble weights w_k that sum to one:

$$\min_{w \in \Delta_K} \left\{ \frac{1}{n} \sum_{i=1}^n \left(y_i - \sum_{k=1}^K w_k p_{ik} \right)^2 + \lambda_w \sum_{k=1}^K \left(w_k - \frac{1}{K} \right)^2 \right\}, \quad (27)$$

where $\Delta_K = \{w : w_k \geq 0, \sum_k w_k = 1\}$. The reference implementation uses $\lambda_w = 0.01$. The small penalty discourages unstable weight concentration around a single learner while retaining a convex ensemble. A still later validation stage calibrates the combined ensemble.

Beginner checkpoint. Think of the three models as three students answering the same probability question. FinCompass does not simply average them. It first checks how each student's probability estimates behave, then learns how much weight to give each student on a later period, then calibrates the combined answer on a still later period. The final exam—the locked test—is not opened until those choices are finished.

6.8 Locked-Test Discipline

If the test result influences the model you choose, the test is no longer an independent test. Repeatedly looking at test performance and changing the model is a form of data snooping and can produce attractive but fragile results [White, 2000, Bailey et al., 2017, Harvey et al., 2016].

FinCompass therefore uses chronological train/validation/test separation, purging, embargo and a locked final test. Section 8 derives these controls.

Implementation note. The frozen-anchor feature construction is in `forecasting/features.py`; Bayesian logistic inference is in `forecasting/bayesian.py`; the calibrated three-component ensemble, validation-stage separation and locked-test gate are in `forecasting/model.py`. The strict reference profile samples approximately every 21 trading sessions, forecasts 252 sessions ahead and uses an explicit 252-business-day embargo around relevant boundaries, implemented with `pandas BDay(252)`.

7 Realtime Adaptation

A frozen model has a strength and a weakness. Its strength is that its validation remains interpretable because the model does not change. Its weakness is that new information may arrive before the next retraining cycle. FinCompass addresses this with an *anchor plus adaptive residual*: the anchor remains frozen, while a smaller online model is allowed to propose a bounded correction in log-odds space.

7.1 Live Feature Vector

The current adaptive residual uses the following ordered features:

1. stock return from the previous session close to the latest available price;
2. the corresponding stock-minus-benchmark return;
3. a z-score of the most recent stock hourly return within the latest 20 hourly-return observations;
4. a z-score of the most recent stock-minus-benchmark hourly return within the latest 20 aligned hourly-return observations;
5. a z-score of the most recent hourly volume within the latest 20 hourly-volume observations;
6. the high-low range across up to the most recent eight hourly bars, divided by the latest stock price;
7. SEC-filing freshness;
8. freshness-weighted indicator for the 8-K/6-K family;
9. freshness-weighted indicator for 10-Q;
10. freshness-weighted indicator for the 10-K family;
11. yield-curve change;
12. high-yield spread change;
13. macro-data freshness.

Call this vector $z_t \in \mathbb{R}^{13}$.

7.2 Frozen Anchor

Let p_t^A be the anchor probability. Convert it to log-odds:

$$\ell_t^A = \text{logit}(p_t^A). \quad (28)$$

7.3 Adaptive Log-Odds Shift

Let online coefficient mean be m_t . The raw residual is

$$\Delta_t^{\text{raw}} = z_t^\top m_t. \quad (29)$$

It is then bounded:

$$\Delta_t = \text{clip}(\Delta_t^{\text{raw}}, -\delta_{\max}, \delta_{\max}). \quad (30)$$

The Balanced profile uses $\delta_{\max} = 0.75$ log-odds units.

The candidate probability is

$$p_t^C = \text{clip}_p\left(\text{sigmoid}(\ell_t^A + \Delta_t)\right). \quad (31)$$

The probability clip in the Balanced profile is 0.01, preventing exact 0 or 1 outputs.

Worked example. Take anchor probability $p_t^A = 0.58$, so $\ell_t^A \approx 0.323$. Suppose the 13 features and current coefficient mean produce raw residual $\Delta_t^{\text{raw}} = 0.33$. Since 0.33 is inside $[-0.75, 0.75]$, no clipping is needed.

$$p_t^C = \text{sigmoid}(0.323 + 0.33) = \text{sigmoid}(0.653) \approx 0.658.$$

The candidate moves to about 65.8%.

Now suppose an extreme event produces raw shift +1.40. The Balanced profile clips it to +0.75:

$$p_t^C = \text{sigmoid}(0.323 + 0.75) = \text{sigmoid}(1.073) \approx 0.745.$$

The bound prevents the realtime layer from overpowering the validated anchor with an arbitrarily large single-event shift.

In odds language, adding Δ multiplies the anchor odds by e^Δ . With $\Delta = 0.75$, the maximum odds multiplier is about $e^{0.75} \approx 2.12$; with $\Delta = -0.75$, it is about 0.47. This is substantial but deliberately bounded.

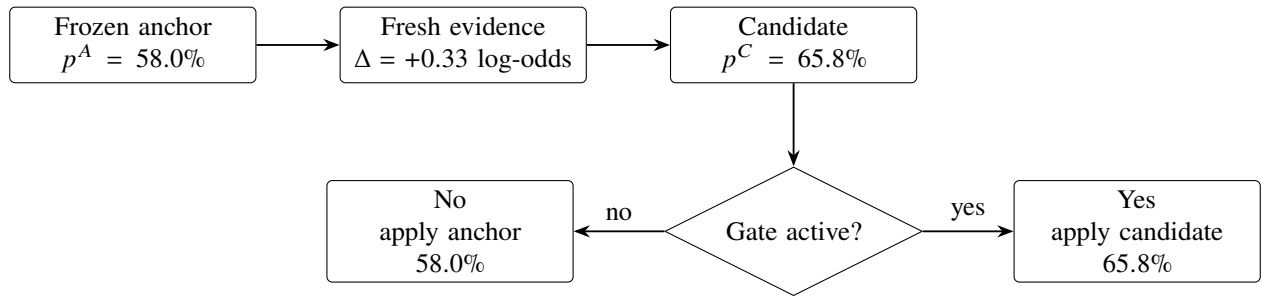


Figure 4: A realtime candidate can move immediately while the applied live probability remains fail-closed. FinCompass uses the candidate only after temporal-breadth, calibration/performance, freshness and drift controls pass.

7.4 Residual Uncertainty

Let online coefficient covariance be P_t . Before log-odds clipping, the local Gaussian model assigns the raw residual

$$\Delta_t^{\text{raw}} = z_t^\top m_t \quad (32)$$

variance

$$\text{Var}(\Delta_t^{\text{raw}}) = z_t^\top P_t z_t. \quad (33)$$

The software reports

$$s_t = \sqrt{z_t^\top P_t z_t} \quad (34)$$

as a coefficient-state uncertainty diagnostic. It is not the standard deviation of the clipped candidate probability and it is not a complete predictive uncertainty interval for market returns.

7.5 Evidence Freshness

New information should not remain “new” forever. An event of age a hours can be assigned an exponential freshness factor

$$f(a) = 2^{-a/h_{1/2}}, \quad (35)$$

where $h_{1/2}$ is a half-life. For SEC filing events, the Balanced profile uses a 36-hour half-life. The macro freshness term uses four times that value, or 144 hours. Market features are not half-life-decayed; they are suppressed when the market provider has not been successfully verified within its configured staleness limit. For the SEC half-life:

$$f(0) = 1, \quad f(36) = 0.5, \quad f(72) = 0.25.$$

FinCompass also distinguishes event age from provider-verification age. An old filing can legitimately remain the latest filing, but if the SEC source has not been checked successfully within the configured safety window, the system cannot know whether a newer filing was missed. Source-related adaptive features are therefore suppressed when verification becomes too stale.

7.6 Predict-Then-Learn Sequence

The most important online-learning rule is temporal:

$$\boxed{\text{predict first; update coefficients only after the original forward label matures.}} \quad (36)$$

If the horizon is 252 common trading sessions, a forecast made today does not provide a true label tomorrow. FinCompass stores a pending observation with its original anchor, feature vector, benchmark contract and exact target endpoint. When the H th common future session exists, the system evaluates the prediction at that endpoint.

7.7 Sequential Gaussian Update

Before each update, the covariance is discounted and process noise is added:

$$P_t^- = \frac{P_{t-1}}{\lambda} + qI, \quad (37)$$

where λ is the forgetting factor and q is process noise. The Balanced profile uses $\lambda = 0.997$ and $q = 0.0005$.

For current matured observation (z, y) and pre-update candidate probability p , the Bernoulli curvature is $p(1 - p)$. The implementation applies a numerical floor,

$$w = \max\{p(1 - p), 10^{-6}\}. \quad (38)$$

Let

$$v = P_t^- z, \quad d = 1 + wz^T v.$$

The rank-one covariance update is

$$P_t = P_t^- - \frac{w}{d} vv^T, \quad (39)$$

and the coefficient-mean update is

$$m_t = m_{t-1} + \frac{v}{d}(y - p). \quad (40)$$

These equations form a computationally efficient local Gaussian (Laplace/quasi-Newton) approximation to a sequential logistic update. They avoid recomputing a full matrix inverse for every matured label. They are not an exact conjugate Bayesian posterior update. In particular, when the deployed log-odds or probability clipping is active, the update uses the local logistic curvature associated with the issued candidate rather than the derivative of the clipped deployment map.

Worked example. Use a one-feature toy version so the arithmetic fits on one line. Suppose $P^- = 0.50$, feature $z = 1.2$, predicted probability $p = 0.60$, label $y = 1$, and current mean $m = 0.10$.

First,

$$w = p(1 - p) = 0.60(0.40) = 0.24.$$

Then

$$v = P^- z = 0.50(1.2) = 0.60.$$

The denominator is

$$d = 1 + 0.24(1.2)(0.60) = 1 + 0.1728 = 1.1728.$$

New covariance:

$$P = 0.50 - \frac{0.24}{1.1728}(0.60)^2 \approx 0.50 - 0.0737 \approx 0.4263.$$

New mean:

$$m' = 0.10 + \frac{0.60}{1.1728}(1 - 0.60) \approx 0.10 + 0.2046 \approx 0.3046.$$

Because the event occurred and the feature was positive, the coefficient moves upward. The covariance shrinks because the model learned from an additional resolved outcome.

7.8 Adaptive Activation Gate

The online model is evaluated against the anchor using matured outcomes. The Balanced profile requires, among other controls:

- at least 120 matured observations;
- at least 60 distinct observation dates;
- at least 180 calendar days of span;
- adaptive Brier score no worse than the anchor under the configured regret threshold;
- adaptive log loss no worse than the anchor;
- expected calibration error no greater than 0.08;
- no active drift alert.

Let A_t indicate that the frozen anchor is eligible for live use. Let G_t denote the statistical adaptive gate, together with adaptive application being enabled and a recently verified market source. The applied probability is

$$p_t^{\text{live}} = \begin{cases} \text{unavailable,} & A_t = 0, \\ p_t^C, & A_t = 1 \text{ and } G_t = 1, \\ p_t^A, & A_t = 1 \text{ and } G_t = 0. \end{cases} \quad (41)$$

Stale SEC or macro verification does not by itself force $G_t = 0$; instead, the associated source-dependent features are set to zero until successful verification resumes. This distinction matches the implementation's fail-closed source semantics.

Beginner checkpoint. The system is allowed to say, “I saw new information and my candidate changed, but I have not yet earned the right to use the adaptive answer.” That is a feature, not a bug.

Implementation note. The feature order, bounded residual, posterior shift uncertainty, date-balanced calibration, drift state and rank-one update are implemented in `realtime/adaptive.py`. Event acquisition/freshness and exact label resolution are handled by the realtime service/provider layer. Adaptive state identity includes the base-model identity, realtime-engine version and a fingerprint of adaptive settings so incompatible configurations cannot silently share one posterior lineage.

8 Validation and Calibration

A probability is useful only if it is evaluated as a probability. Accuracy alone is not enough. A model that labels everything 0.51 can be directionally correct more often than chance in some sample and still be uselessly calibrated. FinCompass therefore evaluates proper probability scores, discrimination, calibration, uncertainty and temporal stability.

8.1 Brier Score

For binary label $y_i \in \{0, 1\}$ and probability p_i ,

$$\text{Brier} = \frac{1}{n} \sum_{i=1}^n (p_i - y_i)^2. \quad (42)$$

Lower is better.

Worked example. Three forecasts are 0.8, 0.6 and 0.3. Outcomes are 1, 0 and 0.

$$(0.8 - 1)^2 = 0.04, \quad (0.6 - 0)^2 = 0.36, \quad (0.3 - 0)^2 = 0.09.$$

Average:

$$\frac{0.04 + 0.36 + 0.09}{3} = 0.1633.$$

The confident wrong 0.6 forecast is penalized more than the correct 0.8 or 0.3 forecasts.

Brier skill compares a model with a reference:

$$\text{BrierSkill} = 1 - \frac{\text{Brier}_{\text{model}}}{\text{Brier}_{\text{ref}}}. \quad (43)$$

Positive values mean improvement over the reference. For the locked-test evaluation, the reference probability is the event rate estimated from the development data, not the event rate recomputed from the locked test. This avoids using test outcomes to define the benchmark forecast.

8.2 Log Loss

Log loss is

$$\text{LogLoss} = -\frac{1}{n} \sum_i [y_i \ln p_i + (1 - y_i) \ln(1 - p_i)]. \quad (44)$$

It strongly punishes confident wrong probabilities. This discourages a model from producing near-0 or near-1 probabilities without strong evidence.

8.3 ROC AUC

ROC AUC measures whether positive cases tend to receive higher scores than negative cases. AUC of 0.5 is roughly random ranking; 1.0 is perfect ranking. AUC says little by itself about whether a stated 70% probability actually behaves like 70%. That is why calibration is evaluated separately.

8.4 Expected Calibration Error

Partition forecasts into probability bins. For bin b , compare average predicted probability $\bar{p}_b$ with observed event frequency $\bar{y}_b$. A common ECE is

$$\text{ECE} = \sum_b \omega_b |\bar{p}_b - \bar{y}_b|. \quad (45)$$

In the realtime gate, FinCompass gives each *observation date* equal total weight. If a date has many tickers, each ticker receives only a fraction of that date's mass. This prevents one large same-day cross section from dominating the calibration statistic.

Worked example. Suppose Day 1 contains 100 stock observations while Day 2 contains only 10. A row-weighted ECE gives Day 1 ten times as much influence. A date-balanced ECE gives Day 1 total mass 1/2 and Day 2 total mass 1/2. Each Day-1 row receives weight $1/(2 \times 100) = 0.005$; each Day-2 row receives $1/(2 \times 10) = 0.05$.

8.5 Calibration Slope

A calibration model can compare observed outcomes with forecast log-odds. FinCompass clips probabilities to $[10^{-6}, 1 - 10^{-6}]$ for this diagnostic and fits an approximately unregularized logistic regression (scikit-learn logistic regression with $C = 10^6$). A slope near 1 is desirable. A slope much less than 1 often indicates predictions that are too extreme; a slope much greater than 1 can indicate under-dispersion or other mismatch. The strict reference configuration accepts a broad slope interval rather than pretending exact 1.0 is required in finite noisy data.

8.6 Purge and Embargo

A 252-session target means nearby observations can share much of the same future outcome window. If a training label ends after a validation period begins, information from the future validation regime can leak into training through the label.

Let observation i begin at t_i and end at target date u_i . For a validation/test segment beginning at T , a purged training observation must satisfy

$$u_i < T. \quad (46)$$

The field called an embargo also creates a pre-boundary exclusion gap. With configured value E , the implementation uses pandas business-day offsets and requires

$$t_i < T - \text{BDay}(E), \quad (47)$$

where $\text{BDay}(E)$ denotes the business-day offset used by the software. The configuration field retains the historical name `embargo_trading_days`, but its executable semantics are business days.

FinCompass applies purge/embargo not only between train and test but between its three validation roles: component calibration, ensemble stacking and final ensemble calibration.

8.7 Moving Date-Block Bootstrap

Cross-sectional stocks on the same day share market conditions. Long-horizon targets on nearby dates also overlap. Treating all rows as independent can make confidence intervals artificially narrow.

The validation layer therefore resamples dates in moving blocks and carries the cross section attached to each selected date. A natural automatic block length is related to

$$B \approx \left\lceil \frac{H}{s} \right\rceil, \quad (48)$$

where H is forecast horizon and s is sampling step. With $H = 252$ and $s = 21$, $B = 12$ observation dates.

8.8 Walk-Forward Validation

A single test period can be lucky or unlucky. Purged walk-forward folds ask whether the model's relative performance remains acceptable across several chronological segments. FinCompass requires a minimum fraction of positive walk-forward Brier-skill folds under the configured profile.

8.9 Validation Tiers

The forecasting framework distinguishes four states:

- fixture_only** Synthetic or deterministic regression evidence proves software/statistical mechanics only.
- rejected** The model failed one or more required statistical/governance gates.

validated_research

A real-data model passed the defined research validation protocol but may still lack some market-grade data controls.

validated_market

In addition to statistical gates, the dataset documents point-in-time features, historical-universe/survivorship controls, delistings and corporate-action-adjusted prices to the required standard.

The bundled reference fixtures remain `fixture_only`. This is intentional.

8.10 Adaptive Drift Monitoring

For each matured observation, define loss difference

$$d_t = (p_t^C - y_t)^2 - (p_t^A - y_t)^2. \quad (49)$$

Positive d_t means the adaptive candidate was worse than the anchor on that observation. FinCompass maintains an exponentially weighted mean and a smoothed squared-deviation scale of this difference and triggers a drift alert when deterioration exceeds a configured control allowance after enough samples. The second quantity is an operational dispersion statistic, not an unbiased sample-variance estimator and not a formal hypothesis-test variance.

A drift alert disables adaptive application; it does not destroy the frozen anchor.

Important limitation. Passing a validation gate is not proof that a relationship will continue forever. It is evidence that the model met a specified standard on specified historical data under specified assumptions. Market regimes, data quality and company behavior can change.

9 Wyckoff Market Structure

Richard D. Wyckoff's early work emphasized reading price, volume, demand and supply rather than relying on a single indicator [Wyckoff, 1910]. Modern presentations of the method commonly organize trading ranges into phases and events such as selling climax, spring, sign of strength, buying climax and upthrust [StockCharts ChartSchool, 2026, Wyckoff Analytics, 2026]. The vocabulary is useful because it turns an abstract chart into a sequence of questions about who appears to be winning the contest between demand and supply.

The framework should nevertheless be treated as a descriptive hypothesis, not a natural law. A spring is not required in every accumulation. A UTAD is not required in every distribution. Labels are often clearest in hindsight. FinCompass therefore uses Wyckoff language in this manual as an interpretation layer, not as a bundled validated forecast model.

9.1 Four Broad Market States

A simplified price cycle is:

$$\text{Accumulation} \rightarrow \text{Markup} \rightarrow \text{Distribution} \rightarrow \text{Markdown}. \quad (50)$$

Accumulation A prior decline loses force and price spends time in a range while supply may be absorbed. The key question is whether downward attempts are becoming less effective and whether demand begins to show itself.

Markup Price leaves the range and establishes a sustained rising structure. The key question shifts from "is the bottom forming?" to "is the trend still healthy?"

Distribution A mature advance stops making easy progress and supply may be entering the market. The key question is whether apparent breakouts are being rejected and whether rallies are becoming less effective.

Markdown Supply dominates and price trends lower. The key mistake to avoid is assuming that every lower price is automatically a bargain.

Beginner checkpoint. Think of a school tug-of-war. Price tells you which team moved the rope; volume gives some information about how much effort was used. Wyckoff analysis asks whether the same amount of effort is producing the same result as before. If sellers need enormous effort to push price only slightly lower, demand may be absorbing supply. If buyers need enormous effort to make almost no upward progress, supply may be absorbing demand.

9.2 Accumulation Phases A–E

Figure 5 is a schematic, not a template that every security must follow.

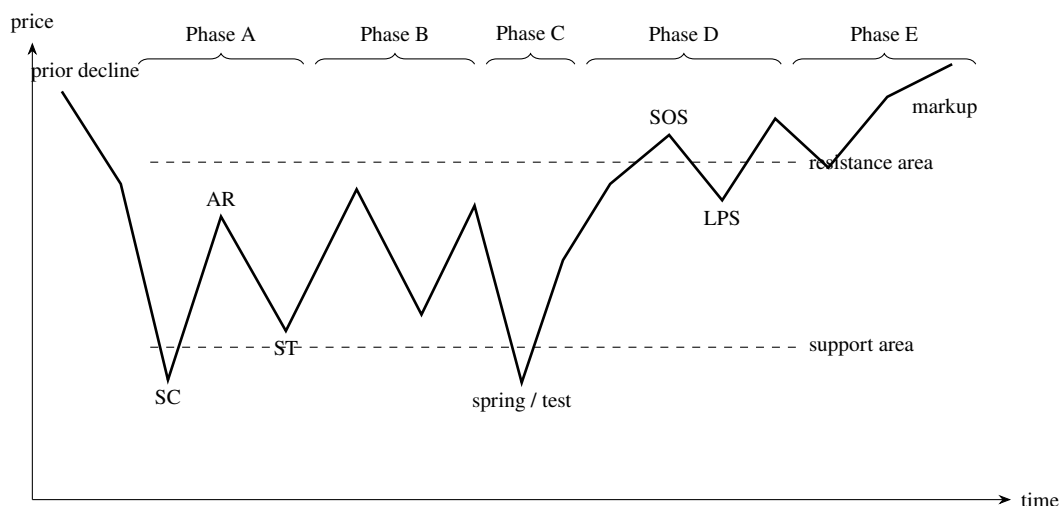


Figure 5: Illustrative Wyckoff accumulation schematic. SC = selling climax, AR = automatic rally, ST = secondary test, SOS = sign of strength, LPS = last point of support. A spring is optional, not mandatory. The figure is schematic and not to scale.

Phase A: the decline stops behaving normally. A selling climax (SC) may show intense selling followed by a sharp response. An automatic rally (AR) establishes an initial upper boundary. A secondary test (ST) revisits the lower region. The important idea is not the label but the change in behavior: a trend that was previously easy to continue begins to stall.

Phase B: the range builds a cause. Price moves back and forth while supply and demand are tested. This phase can be long and frustrating. A beginner's common error is to interpret every mid-range bounce as confirmation that accumulation is complete.

Phase C: a decisive test may occur. A spring or shakeout can briefly break below support and then recover. A successful test can suggest that supply below the range has been absorbed, but springs can fail. Some accumulations proceed without one.

Phase D: demand begins to prove itself. A sign of strength (SOS) pushes price toward or above the upper range. Pullbacks called last points of support (LPS) are often watched for evidence that former resistance is becoming support and that selling pressure is smaller than during earlier reactions.

Phase E: markup becomes visible. Price leaves the old range. At this stage the question is usually no longer whether to guess a bottom. It is whether the trend remains healthy and whether valuation, business evidence and position size still justify holding or adding.

9.3 Distribution Phases

Distribution is often presented as the broad mirror of accumulation, but real markets are not geometrically symmetric.

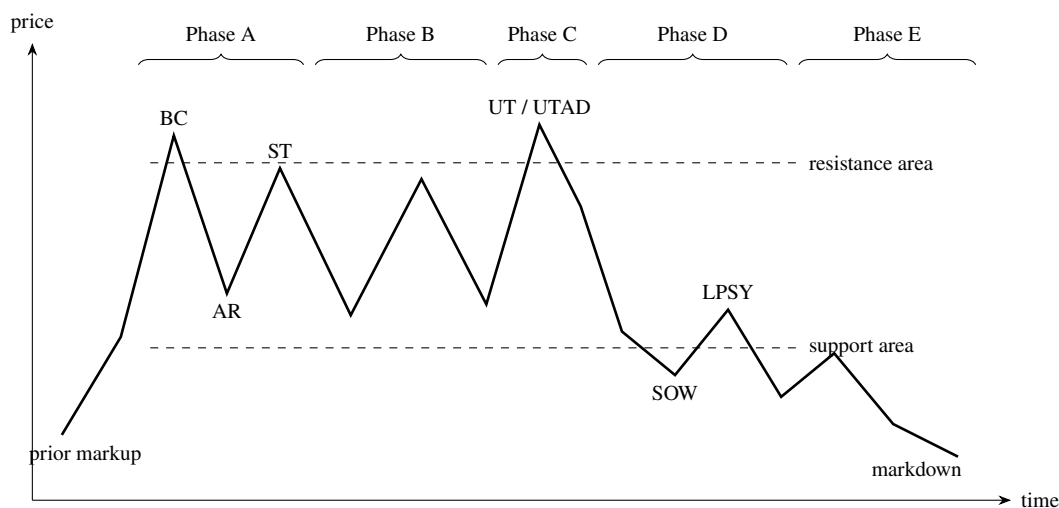


Figure 6: Illustrative Wyckoff distribution schematic. BC = buying climax, AR = automatic reaction, ST = secondary test, UT/UTAD = upthrust/upthrust after distribution, SOW = sign of weakness, LPSY = last point of supply. The figure is schematic and not to scale.

A buying climax (BC) and automatic reaction may establish a range. A later upthrust (UT) or UTAD may briefly break above resistance and then fail. In Phase D, a sign of weakness (SOW) and weak rally toward a last point of supply (LPSY) can indicate that supply is gaining control. Phase E is markdown.

9.4 Three Wyckoff Principles

Modern teaching often summarizes Wyckoff with three “laws.” It is safer to interpret them as analytical heuristics:

1. **Supply and demand.** Price tends to rise when aggressive demand persistently overwhelms available supply and fall when supply overwhelms demand.
2. **Cause and effect.** Time spent building a trading range can precede a larger directional move. The magnitude is not mechanically knowable from the range alone.
3. **Effort versus result.** Volume/transaction effort should be interpreted together with resulting price progress. Large effort with little progress can be informative about opposing liquidity.

These ideas are useful because they force the analyst to compare multiple observations rather than treat one candlestick as destiny.

9.5 Role in FinCompass

FinCompass already supplies evidence quality, valuation/regime context, calibrated forecasting infrastructure and realtime event tracking. Wyckoff can add a human-readable description of *where price appears to be within a trading structure*. It can therefore help answer questions such as:

- Is a falling price still in an obvious markdown, or is downward progress becoming less effective?
- Is a supposed “cheap” stock showing any evidence that demand has returned?
- Is an established markup still orderly, or are repeated breakouts failing?
- Does a trim candidate have both portfolio/fundamental reasons and distribution-like price behavior?

It cannot, without separate validation, convert those observations into a calibrated probability of future excess return.

10 Quantifying Wyckoff Structure

A useful teaching framework should not depend entirely on labels drawn by eye. This section shows simple measurements that make Wyckoff-style observations reproducible. These are **descriptive features**, not validated return probabilities in the current implementation.

10.1 Range Position

For a chosen trading range with support L and resistance H , with $H > L$,

$$r_t = \frac{C_t - L}{H - L}, \quad (51)$$

where C_t is the close. Values near 0 are near support; values near 1 are near resistance.

A low r_t is not bullish by itself. In a markdown, price can remain near the lower boundary while the boundary keeps moving downward.

10.2 Relative Volume

Let V_t be current volume and $\tilde{V}_n$ the median volume over the prior n sessions. Define

$$RV_t = \frac{V_t}{\tilde{V}_n}. \quad (52)$$

If $RV_t = 1.8$, volume is 80% above its reference median.

Worked example. Median 20-day volume is 2.0 million shares. Today volume is 3.6 million.

$$RV_t = \frac{3.6}{2.0} = 1.8.$$

That is high effort. The next question is what price accomplished with that effort.

10.3 Relative Price Spread

Let daily high and low be H_t and L_t . Define spread

$$S_t = H_t - L_t.$$

With median recent spread $\tilde{S}_n$,

$$RS_t = \frac{S_t}{\tilde{S}_n}. \quad (53)$$

A session with $RV_t = 2.0$ but $RS_t = 0.6$ has high trading effort and relatively little range expansion. That can be consistent with absorption, although direction, location and subsequent confirmation still matter.

10.4 Close Location

Define

$$CLV_t = \frac{C_t - L_t}{H_t - L_t}, \quad (54)$$

when $H_t > L_t$. A value near 1 means the close finished near the day's high; near 0 means near the day's low.

10.5 Benchmark-Relative Momentum

For n sessions,

$$M_{t,n}^{\text{rel}} = R_{\text{stock},t,n} - R_{\text{benchmark},t,n}. \quad (55)$$

If a stock rises 4% while the benchmark rises 7%, relative momentum is -3% . In a possible accumulation, improving relative behavior can be more informative than an isolated positive day.

10.6 Support Rejection

Suppose pre-existing support is L . A simple rejection depth is

$$d_t = \frac{L - L_t}{L}, \quad (56)$$

for a low $L_t < L$. A spring-like *candidate* can be described with conditions such as

$$I_{\text{reject}} = \mathbb{I}\{L_t < L\} \mathbb{I}\{C_t > L\}. \quad (57)$$

This says only that price traded below support and closed back above it. It does not say the event will succeed.

Worked example. Range support is \$50. During the session price trades to \$48.50 but closes at \$51.20.

$$d = \frac{50 - 48.5}{50} = 0.03 = 3\%.$$

The low broke support by 3%, but the close recovered above support. That satisfies the simple rejection condition. Next examine volume, spread, subsequent test behavior, fundamentals and broader risk before assigning an interpretation.

10.7 Breakout and Sign of Strength

For resistance H , a simple breakout magnitude is

$$b_t = \frac{C_t - H}{H}. \quad (58)$$

A positive b_t means the close is above the defined resistance. A stronger SOS hypothesis would normally ask for additional evidence: expanding spread, supportive volume, improving relative strength and follow-through rather than a one-day close alone.

10.8 Retest and Last Point of Support

After breakout level H , let a pullback low be L_t^{pull} . Define retest depth

$$q_t = \frac{L_t^{\text{pull}} - H}{H}. \quad (59)$$

A small nonnegative q_t means the pullback stayed above the old resistance. A slightly negative q_t followed by recovery may still be acceptable depending on volatility. A descriptive LPS candidate becomes stronger if pullback volume/spread are lower than on the preceding advance.

10.9 Upthrust and Failed Breakout

The mirror condition above resistance is

$$I_{\text{upthrust}} = \mathbb{I}\{H_t > H\} \mathbb{I}\{C_t < H\}. \quad (60)$$

A high above resistance followed by a close back inside the range is a measurable rejection. The label UT or UTAD still requires structural context; the equation only identifies the rejection geometry.

10.10 Effort vs. Result

Rather than invent a mysterious oscillator, compare two normalized numbers:

$$E_t = RV_t, \quad R_t = RS_t. \quad (61)$$

If E_t is very high but R_t is modest, price is absorbing substantial trading effort with limited range result. The sign and close location determine whether the absorption appears more consistent with demand or supply.

A teaching-only ratio can be written

$$A_t = \frac{RV_t}{\max(RS_t, \epsilon)}, \quad (62)$$

but A_t is *not* a standard Wyckoff statistic and should not be assigned universal thresholds without validation.

10.11 Phase Evidence

Let $g_{kj}(x_t) \in [0, 1]$ represent the degree to which measurement j supports phase hypothesis k . A transparent descriptive evidence index is

$$E_k(t) = \frac{\sum_j a_{kj} g_{kj}(x_t)}{\sum_j a_{kj}}, \quad (63)$$

where $a_{kj} \geq 0$ are declared weights.

The crucial word is *evidence*. Unless E_k has been trained/calibrated against a point-in-time phase definition and then validated out of sample, $E_k = 0.8$ must not be called “80% probability of accumulation.”

10.12 Probabilistic Phase Model

A natural research extension is a hidden-state model in which phase C_t is not directly observed. With transition matrix T and phase-specific feature likelihood $L_k(x_t)$, a filtered phase probability could have the form

$$\pi_t(k) \propto L_k(x_t) \sum_j T_{jk} \pi_{t-1}(j). \quad (64)$$

This would formalize persistence and plausible phase transitions. However, the hard problem is not writing Equation (64); it is obtaining defensible point-in-time phase labels and demonstrating calibration. The current manual therefore keeps this as a research extension rather than a product claim.

10.13 Chart-Review Worksheet

For each suspected trading range, record:

Quantity	Calculation	Question
Range position	$(C - L)/(H - L)$	Is price near support, middle, or resistance?
Relative volume	$V/\tilde{V}$	Is current trading effort unusual?
Relative spread	$(H_t - L_t)/\tilde{S}$	Is price result large or small relative to normal?
Close location	$(C - L_t)/(H_t - L_t)$	Did the session close near its high or low?
Relative momentum	$R_{stock} - R_{bench}$	Is the security improving or weakening versus its benchmark?
Support rejection	Eq. (57)	Did a downside break recover into the range?
Resistance rejection	$I_{upthrust}$	Did an upside break fail back into the range?
Follow-through	multi-session	Did the next few sessions confirm or invalidate the hypothesis?

This worksheet creates an audit trail. A reader can later see whether the original conclusion was based on observable conditions or on hindsight.

11 Action Compass

The word “action” is where a research tool can become dangerous if it jumps directly from a model output to a trading instruction. FinCompass does not know the user’s income, debts, emergency fund, tax situation, liquidity needs, concentration, employer exposure or psychological ability to tolerate losses. The safest design is therefore a sequence of gates and decision-support categories rather than a universal buy/sell score.

11.1 Decision Inputs

Let the decision information state be

$$\mathcal{D}_t = \{E_t, P_t, U_t, W_t, R_t, K_t\}, \quad (65)$$

where

- E_t = business/evidence state;
- P_t = validated forecast state, if one is live-eligible;
- U_t = uncertainty, calibration and freshness state;
- W_t = Wyckoff/market-structure description;
- R_t = security-specific risk/thesis state;
- K_t = portfolio constraints such as position size, diversification and available capital.

The manual’s Action Compass is a rule map

$$A_t = \phi(\mathcal{D}_t), \quad (66)$$

not a statistically calibrated return model. The map produces categories such as *watch*, *routine DCA*, *staged accumulation candidate*, *hold*, *hold/no add*, *trim candidate* and *reduce-risk candidate*.

11.2 Thesis Gate

Before adding to a single company, ask whether the original business thesis still exists. Examples of thesis-break evidence include severe balance-sheet deterioration, structural loss of competitiveness, accounting uncertainty, permanent impairment of a key market, or an adverse event that invalidates the reason the security was owned.

Represent the result as

$$G_{\text{thesis}} \in \{0, 1\}. \quad (67)$$

If $G_{\text{thesis}} = 0$, a lower price is not a reason to average down. The appropriate category is review/reduce risk, not automatic DCA.

11.3 Evidence-Quality Gate

A high score with very low coverage is not equivalent to a high score with broad evidence. A generic evidence gate can be written

$$G_{\text{evidence}} = \mathbb{I}\{C_t \geq C_{\min}\} \mathbb{I}\{\text{no unresolved critical data-quality flag}\}. \quad (68)$$

The threshold $C_{\min}$ is a user governance choice, not a universal investment law.

11.4 Forecast-Governance Gate

If no real model has earned a live validation tier, do not substitute a fixture probability. Define

$$G_{\text{forecast}} = \mathbb{I}\{\text{model tier is live-eligible}\} \mathbb{I}\{\text{target contract matches the question}\}. \quad (69)$$

When this gate is zero, the user can still perform evidence research and chart analysis, but should not describe a synthetic fixture output as an empirical market probability.

11.5 Market-Structure Gate

Wyckoff structure is represented qualitatively or by transparent descriptive evidence. For example:

Accumulation candidate

Support tests become less effective to the downside; rejection/recovery or SOS/LPS behavior appears; relative behavior improves.

Markup

Higher highs/higher lows or equivalent trend structure is established; pullbacks remain controlled.

Distribution candidate

Upward progress becomes difficult; failed breakouts, SOW/LPSY behavior or deterioration in relative strength appears.

Markdown

Lower lows/lower highs or equivalent downward trend remains dominant.

This gate cannot override a broken thesis.

11.6 Portfolio-Risk Gate

Even a favorable security can be inappropriate to add if it is already too large in the portfolio. Define a user-chosen maximum position fraction $k_{\max}$. If current position fraction is k_t , then

$$G_{\text{capacity}} = \mathbb{I}\{k_t < k_{\max}\}. \quad (70)$$

A similar gate can be applied to sector concentration, liquidity and cash-reserve requirements.

Every arrow is conditional. Thesis, evidence quality, portfolio capacity and validated-forecast gates must still pass. Wyckoff structure alone never authorizes a trade.

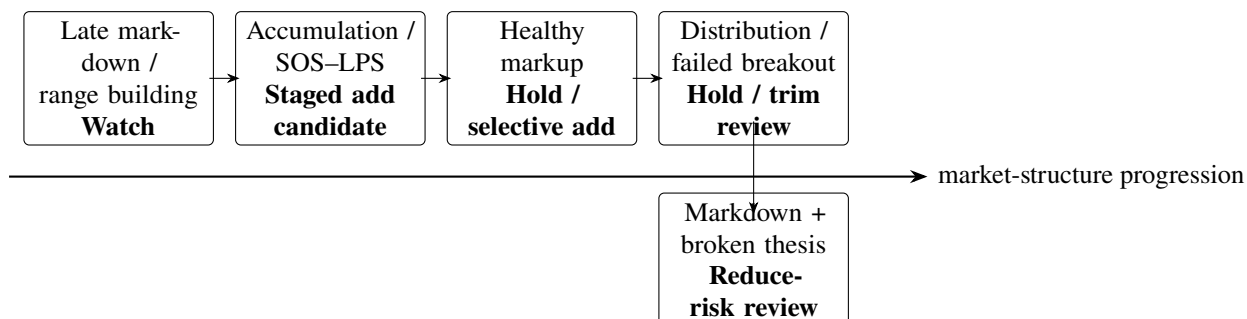


Figure 7: Educational market-structure-to-action map. It helps a beginner organize questions, not predict turning points. The same chart structure can imply a different action when the business thesis, valuation, uncertainty or portfolio concentration differs.

11.7 Action categories

Category	Typical conditions	What it means
Watch	Evidence incomplete, structure uncertain, or forecast gate unavailable	Observe; gather more evidence rather than force a position.
Routine DCA	Long-horizon diversified plan remains intact	Continue planned periodic contributions; do not require a market-timing signal.
Staged accumulation candidate	Thesis intact, evidence adequate, risk capacity available, accumulation/SOS-LPS structure plausible; forecast supportive if validated	Consider dividing intended capital into tranches rather than treating one chart event as certainty.
Hold	Thesis intact, evidence/forecast constructive, markup healthy, position not excessive	Let an existing position work; avoid unnecessary trading.
Hold / no add	Business still acceptable but valuation, uncertainty, concentration or late-stage structure reduces attractiveness of new capital	Maintain while withholding incremental capital.
Trim candidate	Position concentration high and/or valuation/structure/evidence deteriorating	Consider partial risk reduction rather than assuming all-or-nothing liquidation.
Reduce-risk candidate	Thesis broken, severe deterioration, invalidated structure, or portfolio constraint breached	Reassess the position; averaging down is not the default response.

11.8 Routine DCA and Tactical Staging

Suppose an investor has a planned monthly contribution B . Routine DCA means investing B on the schedule independent of short-term market direction [U.S. Securities and Exchange Commission, Investor.gov, 2026]. A

tactical overlay can be represented for teaching as

$$I_t = \min\{B m_t, C_{\text{remaining}}, C_{\text{capacity}}\}, \quad (71)$$

where m_t is an *illustrative, user-defined* multiplier and the two capital constraints prevent spending unavailable or risk-prohibited cash.

This equation is useful for showing discipline, not for specifying universal multipliers. A conservative policy might permit m_t only in a narrow range around 1. A single-stock strategy might instead predefine four equal tranches and release the next tranche only after declared evidence conditions are met.

Worked example. A student sets aside \$1,000 for a hypothetical single-stock research exercise and decides *before seeing the chart* to use four maximum tranches of \$250. The first tranche is allowed only if the business-thesis and portfolio-capacity gates pass. A second tranche requires a successful support test and improving evidence. A third requires a sign-of-strength/retest structure. The fourth is never automatic; it is held back if valuation becomes demanding or the thesis changes.

The arithmetic is simple:

$$1000/4 = 250 \text{ per maximum tranche.}$$

What matters is that the tranche plan is specified before emotion enters. If the thesis breaks after the first tranche, the remaining \$750 is not forced into the security merely because the price is lower.

11.9 Profit-Taking Rules

A stock being profitable is not, by itself, a reason to sell. A more defensible trim framework asks whether one or more constraints have changed:

- the position has grown beyond a portfolio cap;
- valuation has become extreme relative to the user's framework;
- the calibrated forecast has deteriorated materially;
- distribution-like structure has emerged and persisted;
- the thesis has weakened;
- the investor needs liquidity or rebalancing for reasons outside the stock.

A partial trim can be written as

$$Q_{\text{sell}} = \max\{0, Q_t - Q_{\text{target}}\}, \quad (72)$$

where Q_t is current shares and Q_{target} is the number consistent with the target portfolio weight. This is a portfolio-control equation, not a market forecast.

11.10 Decision Tree

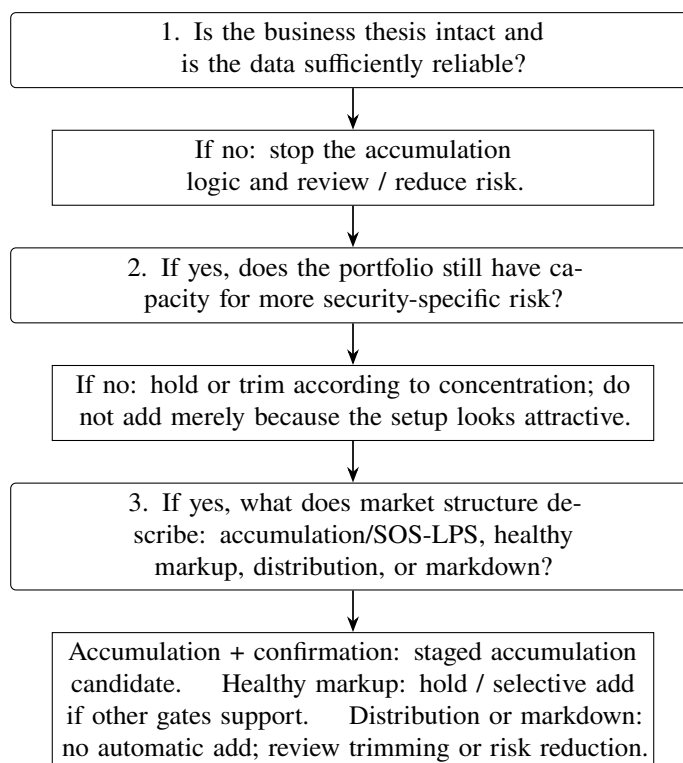


Figure 8: Educational Action Compass. It is a guardrail sequence, not a validated trading model. Personal suitability constraints remain outside FinCompass.

12 Case Studies

Every case in this section is synthetic. The purpose is to show the reasoning process, including situations in which the correct response is to do nothing. The numbers are chosen to make the arithmetic visible; they are not forecasts for any real security.

12.1 Diversified Index DCA

Situation. Maya contributes \$300 each month to a broad, diversified equity ETF as part of a long-horizon savings plan. The market falls 12% over several months. She asks whether she should stop the plan until the market “looks safer.”

Step 1: identify the strategy. This is routine DCA from ongoing income, not a lump sum waiting in cash. The planned monthly contribution is

$$B = \$300.$$

The DCA rule is simply $I_t = B$ each month, provided Maya’s personal financial plan still supports the contribution.

Step 2: apply the Action Compass. A broad ETF has a different failure mode from a single company. There is no single-company thesis to break, although portfolio suitability and liquidity still matter. A Wyckoff markdown label is not enough to stop a long-horizon contribution plan.

Step 3: explain the arithmetic. If the ETF price is \$100, \$300 buys 3 shares. If the price falls to \$75, the same \$300 buys 4 shares. If it rises to \$120, \$300 buys 2.5 shares. This is the mechanical meaning of DCA.

Interpretation. FinCompass can provide context, but the core strategy is contribution discipline, not predicting the exact bottom. Investor.gov describes DCA as investing equal portions at regular intervals through market ups and downs [U.S. Securities and Exchange Commission, Investor.gov, 2026].

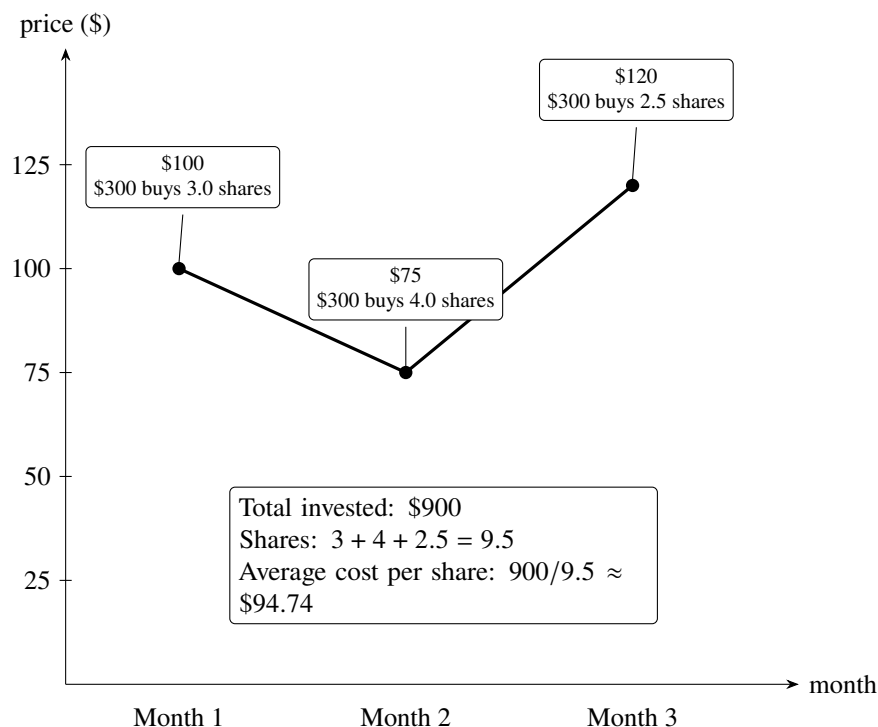


Figure 9: Grade-10 DCA arithmetic with equal dollar contributions. More shares are purchased when price is lower and fewer when price is higher. This does not make a deteriorating single company safe; the thesis and risk gates still matter.

Professor-level note. If Maya instead received \$30,000 in cash today, the question changes. Deliberately holding cash to stage entry has an opportunity cost; historical Vanguard analysis found immediate lump-sum investment outperformed temporary cost averaging roughly two-thirds of the time in its study design [Wimmer and Zorn, 2023]. The optimal choice therefore depends partly on risk aversion and behavior, not only expected return.

12.2 Late-Markdown Accumulation

Situation. Synthetic Company Alpha fell from \$120 to a range between \$70 and \$85 after a broad market correction. FinCompass evidence is 7.7 with 88% coverage. The thesis is intact. No live-eligible forecast is installed. Price recently dipped below \$70 to \$67.90 and closed at \$72 on 1.6 times median volume.

Step 1: measure the rejection. Support $L = 70$, intraday low $L_t = 67.9$, close $C_t = 72$.

$$d = \frac{70 - 67.9}{70} = 0.03 = 3\%.$$

The low broke support by 3% and the close recovered above support. The simple rejection indicator is therefore 1.

Step 2: range position. With resistance $H = 85$,

$$r = \frac{72 - 70}{85 - 70} = \frac{2}{15} \approx 0.133.$$

Price is still near the lower part of the range.

Step 3: volume context. Relative volume is $RV = 1.6$. This shows elevated effort. The analyst must still ask whether the next test occurs on lower selling pressure and whether price begins to show a sign of strength.

Step 4: Action Compass. Thesis gate passes; evidence gate passes; portfolio capacity is available. The Wyckoff interpretation is *possible Phase C rejection*, not confirmed accumulation. Because no live-eligible forecast exists, no probability claim is added.

Action category. Staged accumulation candidate, not “all in.” If the student had preallocated \$1,000 in four \$250 tranches, one tranche could be used under the declared plan, while later tranches require confirmation. If the spring fails and the thesis deteriorates, the remaining cash is preserved.

12.3 False Spring and Thesis Failure

Situation. Synthetic Company Beta has fallen 60%. It forms a range from \$18 to \$24. Price briefly falls to \$17 and closes at \$19, apparently producing a spring-like rejection. But evidence score is only 4.8, coverage is 92%, debt has risen sharply, interest coverage is deteriorating, and a major customer has left.

Step 1: do not let the chart erase the business. The support rejection geometry is real:

$$d = \frac{18 - 17}{18} = 5.56\%.$$

Yet the thesis gate fails because material business evidence has deteriorated.

Step 2: understand the Action Compass.

$$G_{\text{thesis}} = 0.$$

Once this gate is zero, a spring-like chart event cannot restore an accumulation decision by itself.

Action category. Reduce-risk / avoid automatic DCA. A lower price is not synonymous with greater value when the denominator of value—future cash generation—has deteriorated.

Lesson. This case exists because a user manual that only shows successful springs teaches the wrong behavior. False positives are part of the model.

12.4 Markup and Holding

Situation. Company Gamma was accumulated earlier and has risen from \$50 to \$78. It makes higher highs and higher lows. Evidence remains 8.1 with high coverage. Position size remains below the user's declared cap. Valuation is no longer cheap but not extreme. The user is tempted to sell simply because the position is profitable.

Step 1: calculate profit but do not confuse profit with a sell signal.

$$R = \frac{78}{50} - 1 = 56\%.$$

A 56% unrealized gain describes the past. It does not answer whether expected future risk-adjusted return is now poor.

Step 2: structure. The chart is in markup with orderly pullbacks; no repeated failed breakout or SOW/LPSY pattern is visible.

Step 3: portfolio gate. Position remains below cap.

Action category. Hold. The manual treats holding as an active decision. Trading merely to “lock in green” can create taxes, transaction costs and the risk of exiting a strong trend without a replacement thesis.

12.5 Re-Accumulation

Situation. Company Delta advances from \$40 to \$70, then spends eight weeks between \$63 and \$72. It does not return to the old base. Pullbacks occur on lower relative volume; a breakout above \$72 is followed by a controlled retest to \$71.50 and renewed strength.

Step 1: breakout magnitude. If the breakout close is \$74,

$$b = \frac{74 - 72}{72} = 2.78\%.$$

Step 2: retest depth.

$$q = \frac{71.5 - 72}{72} = -0.69\%.$$

The pullback briefly slips 0.69% below the old resistance but recovers. Whether this is acceptable depends on normal volatility; the number itself makes the judgment auditable.

Step 3: Action Compass. If business evidence and portfolio capacity remain supportive, this can be classified as *re-accumulation / hold-selective-add candidate*. It is not as attractive as the original base merely because it is a new range; valuation and concentration have changed.

12.6 Concentration-Driven Trimming

Situation. Company Epsilon has appreciated so much that it now represents 18% of a portfolio whose declared single-position cap is 12%. The business remains strong, but valuation is stretched and price has begun to make less progress on high volume.

Step 1: portfolio mathematics. Portfolio value is \$200,000, so the current position is

$$0.18(200,000) = \$36,000.$$

Target cap is

$$0.12(200,000) = \$24,000.$$

An illustrative rebalance amount is

$$36,000 - 24,000 = \$12,000.$$

Step 2: market structure. High effort with declining upward result is consistent with a distribution concern but is not proof of distribution.

Action category. Trim candidate. Importantly, the reason is not “the stock went up too much.” The portfolio constraint is already sufficient to justify reducing concentration; stretched valuation and distribution-like behavior strengthen the case.

12.7 UTAD and Adverse Realtime Evidence

Situation. Company Zeta trades in a mature range with resistance at \$105. It rallies intraday to \$110 but closes at \$102. The next day an adverse 8-K arrives. A validated anchor probability is 0.61. The realtime residual produces raw shift -0.42 and the adaptive gate is active.

Step 1: upthrust geometry. The high broke resistance by

$$\frac{110 - 105}{105} = 4.76\%,$$

but the close finished back below resistance. This is a measurable failed breakout.

Step 2: adaptive probability. Anchor log-odds:

$$\text{logit}(0.61) = \ln(0.61/0.39) \approx 0.447.$$

Apply residual:

$$0.447 - 0.42 = 0.027.$$

Candidate probability:

$$\text{sigmoid}(0.027) \approx 0.507.$$

Because the gate is active, the applied probability can move from 61% to about 50.7% under this synthetic example.

Action category. For an existing overweight position, **trim/reduce-risk review**. For a new buyer, **watch rather than chase**. The key is agreement across independent layers: failed breakout, negative filing context and deterioration in a validated adaptive probability.

12.8 Value Trap in Markdown

Situation. Company Eta falls from \$90 to \$45. P/E looks lower, and the investor says “it is half price.” But earnings fell 55%, leverage worsened, relative strength remains negative and every bounce makes a lower high.

Step 1: price fall.

$$R = 45/90 - 1 = -50\%.$$

Step 2: earnings deterioration. If earnings per share fell from 6.0 to 2.7,

$$\frac{2.7}{6.0} - 1 = -55\%.$$

The business deteriorated even more than the stock price.

Step 3: valuation intuition. If old P/E was $90/6 = 15$ and new P/E is $45/2.7 = 16.67$, the stock is actually *more expensive on current earnings* despite being 50% lower in price.

Action category. **Avoid automatic averaging down.** Wait for business stabilization and evidence that markdown is losing control.

12.9 High Quality and Demanding Valuation

Situation. Company Theta has Quality 9.0, Durability 8.6 and Safety 8.8, but Valuation 3.4 and Cycle 4.5. A validated anchor probability is only 0.53 with a broad interval around 0.5.

Step 1: understand the apparent contradiction. The company can be excellent while the stock is not an obvious new-money opportunity at the current price.

Step 2: simple evidence-weight intuition. Strong business pillars are partly offset by valuation and context. The exact composite comes from posterior propagation, not a recommendation.

Action category. For a current holder with reasonable position size, **hold / no add** may be more coherent than either “sell because expensive” or “buy because quality is high.” For a new investor, **watch** until price, valuation or forecast changes.

12.10 Candidate Rejected by the Gate

Situation. Company Iota has anchor probability 0.58. A strong earnings release and favorable relative move generate candidate probability 0.66. However, the adaptive state has only 35 unique observation dates and 90 days of span. Balanced settings require at least 60 dates and 180 days.

Step 1: candidate. The candidate 0.66 is a legitimate model calculation.

Step 2: gate.

$$35 < 60, \quad 90 < 180.$$

Two temporal-breadth conditions fail.

Step 3: live output.

$$p^{\text{live}} = p^A = 0.58.$$

Lesson. Realtime information and online model authority are separate. The app may show the candidate for inspection while refusing to apply it.

12.11 ETF DCA vs. Single-Stock Averaging Down

Two investors use the same word “DCA” but face different risks.

Investor A contributes to a diversified ETF with a 20-year horizon. Investor B keeps buying one company whose debt, margins and market share deteriorate. In both cases the arithmetic of buying more shares at lower prices is the same. The economic risk is not.

For a diversified fund, periodic DCA can be a disciplined asset-allocation process. For a single company, repeated purchases concentrate idiosyncratic risk. The thesis gate is therefore much more important.

General lesson. DCA is a funding schedule; it is not a substitute for security selection or risk control.

12.12 Conflicting Wyckoff and Model Signals

Situation. Company Kappa shows a spring-like rejection and improving price/volume behavior, but a validated forecast falls from 0.62 to 0.48 after new point-in-time fundamentals. What should the user do?

Reasoning. Do not average the signals into an invented 55% answer. The Wyckoff layer is descriptive and unvalidated as a return probability. The calibrated model carries a different evidentiary status. The conflict should be displayed, not hidden.

Action category. Watch / gather evidence. A later SOS plus improving fundamentals may resolve the conflict. A failed spring may resolve it the other way. Uncertainty itself is actionable information because it can justify waiting.

13 Failure Modes

The most useful user manual is not a catalogue of success stories. This section lists common ways a user can misuse otherwise reasonable mathematics.

13.1 Evidence Score vs. Return Probability

Wrong interpretation: “FinCompass score 8.4 means an 84% chance the stock will go up.”

Correct interpretation: 8.4 is a score on a 0–10 rubric. A statement such as $\Pr(S \geq 8)$ concerns uncertainty in that score. It is not calibrated to future returns unless a separate forecast model has been trained for that event.

13.2 Synthetic Fixtures vs. Market Validation

A fixture can prove that the code recovers known structure, respects split rules, calculates metrics correctly and fails closed. It cannot prove alpha. If the model tier is `fixture_only`, the correct user response is “software/statistical mechanics validated on the fixture,” not “market model validated.”

13.3 Fixed-Horizon Labeling

If a 252-session forecast is processed 270 sessions later, using the latest price would silently change the event. FinCompass resolves at the exact H th common future trading session.

13.4 Vintage Macroeconomic Data

A modern revised value cannot be inserted into an old observation unless it was available then. Vintage-aware data such as ALFRED exist to preserve historical information states [Federal Reserve Bank of St. Louis, 2026].

13.5 Date-Balanced Online Validation

If 500 tickers are observed on one date and 20 on another, row-weighted metrics let the first market regime dominate. Date-balanced online loss and ECE give each observation date equal total mass.

13.6 Support Breaks and Springs

A spring is a structural interpretation, not the equation $L_t < L$ alone. The range must be meaningful, the recovery must be evaluated, subsequent testing matters, and some genuine accumulations occur without springs.

13.7 Failed Breakouts and Distribution

A single failed breakout can occur in an ongoing uptrend. Distribution is a sequence of evidence, not one candle.

13.8 Averaging Down After Thesis Failure

A lower price can coexist with worse valuation if earnings or cash flow fall faster than the price. Case 8 showed this arithmetically. The thesis gate exists to stop a chart pattern from overriding business deterioration.

13.9 Threshold Overfitting

If a user tries 50 different definitions of “spring,” “cheap,” or “strong” on the same history and keeps the one with the best result, the apparent improvement can be data snooping. Any new probabilistic Wyckoff classifier would need pre-registration, separate development/validation/test data and robustness controls.

13.10 Calibration vs. Certainty

A calibrated 70% forecast is expected to be wrong about 30% of the time in the long run for comparable cases. Calibration makes uncertainty meaningful; it does not eliminate it.

13.11 Portfolio Concentration

A stock can remain a good business while becoming an oversized portfolio risk. The decision to trim can therefore be rational even when the company thesis has not failed.

13.12 Fresh but Unverified Data

FinCompass distinguishes event freshness from provider-verification freshness. If a provider has been unavailable too long, the absence of a new event cannot be interpreted as evidence that nothing happened.

13.13 Settings and Model Identity

Changing horizon from 252 to 63 sessions changes the target. Changing SPY to QQQ changes the benchmark. Changing the realtime learning profile changes posterior dynamics. FinCompass fingerprints settings and binds state lineage to those contracts so one configuration cannot silently inherit another’s validation.

13.14 Complexity vs. Performance

A complicated model can fit noise more efficiently than a simple one. The purpose of the ensemble and Bayesian/adaptive layers is not complexity for its own sake. Each layer has a specific role and a gate that can disable it.

13.15 Scope and Claims

The worked cases are educational. They are designed to show arithmetic and reasoning, including failure. Real decisions require current data, taxes, costs, personal constraints and the possibility that the model is wrong.

14 Mathematical Reference

This section restates the main model in a compact form for mathematically trained readers. It is not required for ordinary use of the software.

14.1 Information and Target Measurability

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $\{\mathcal{F}_t\}$ the filtration generated by information available to FinCompass through time. Every feature used at time t must be $\mathcal{F}_t$ -measurable.

For security i , benchmark b , horizon H and hurdle η ,

$$Y_{i,t}^{(H,\eta)} = \mathbb{I} \left\{ \frac{P_i(\tau_H(t))}{P_i(t)} - \frac{P_b(\tau_H(t))}{P_b(t)} > \eta \right\}. \quad (73)$$

Equivalently, writing returns explicitly,

$$Y_{i,t}^{(H,\eta)} = \mathbb{I} \left\{ R_{i,t}^{(H)} - R_{b,t}^{(H)} > \eta \right\}. \quad (74)$$

The intended forecast is

$$p_{i,t} = \mathbb{P}(Y_{i,t}^{(H,\eta)} = 1 \mid \mathcal{F}_t). \quad (75)$$

14.2 Fractional-Beta Evidence Model

For pillar r and metric scores $q_j \in [0, 1]$,

$$\alpha_r = \alpha_0 + c \sum_{j \in \mathcal{J}_r} w_j q_j, \quad (76)$$

$$\beta_r = \beta_0 + c \sum_{j \in \mathcal{J}_r} w_j (1 - q_j). \quad (77)$$

With $U_r \sim \text{Beta}(\alpha_r, \beta_r)$,

$$\mathbb{E}[U_r] = \frac{\alpha_r}{\alpha_r + \beta_r}, \quad (78)$$

$$\text{Var}(U_r) = \frac{\alpha_r \beta_r}{(\alpha_r + \beta_r)^2 (\alpha_r + \beta_r + 1)}. \quad (79)$$

The pillar score is $S_r = 10\mathbb{E}[U_r]$.

For pillar weights ω_r summing to 1, independent propagation is

$$S^{\text{ind}} = \sum_r \omega_r S_r^{(m)}, \quad (80)$$

where $S_r^{(m)}$ are independent posterior draws. A comonotonic sensitivity scenario sorts each pillar's draws before applying the same weighted sum, approximating perfect positive rank dependence. The reported interval is the envelope of the central intervals from the two scenarios.

14.3 Bayesian Logistic Anchor

Let standardized design row be x_i with intercept. For coefficient vector β ,

$$p_i = \text{sigmoid}(x_i^\top \beta). \quad (81)$$

The Bernoulli log likelihood is

$$\log L(\beta) = \sum_i [y_i \log p_i + (1 - y_i) \log(1 - p_i)] . \quad (82)$$

With Gaussian prior $\beta \sim N(0, \Lambda^{-1})$, the negative log posterior is

$$\mathcal{L}(\beta) = \sum_i \left[\log(1 + e^{x_i^\top \beta}) - y_i x_i^\top \beta \right] + \frac{1}{2} \beta^\top \Lambda \beta. \quad (83)$$

Gradient:

$$\nabla \mathcal{L} = X^\top (p - y) + \Lambda \beta. \quad (84)$$

At MAP $\hat{\beta}$, the implementation uses

$$W = \text{diag}\{\max(p_i(1 - p_i), 10^{-8})\}, \quad H = X^\top W X + \Lambda. \quad (85)$$

The 10^{-8} floor is a numerical safeguard. The Laplace approximation is

$$\beta \mid D \approx N(\hat{\beta}, H^{-1}), \quad (86)$$

with the covariance evaluated numerically using the Moore–Penrose pseudoinverse of H .

14.4 Probability Calibration

Let $p^{\text{raw}} \in (0, 1)$ denote an uncalibrated class probability after numerical clipping to $[10^{-6}, 1 - 10^{-6}]$. The implemented sigmoid calibrator fits logistic regression on the raw log-odds,

$$\hat{p} = \text{sigmoid}[a \logit(p^{\text{raw}}) + b]. \quad (87)$$

The implementation uses weak L_2 regularization ($C = 10^4$). Isotonic calibration instead fits a monotone piecewise-constant map. The calibration observations are chronologically separated from the observations used to choose ensemble weights and from the locked test.

14.5 Constrained Probabilistic Stacking

For calibrated component probabilities p_{ik} , ensemble weights solve

$$\min_{w \in \Delta_K} \left\{ \frac{1}{n} \sum_i \left(y_i - \sum_k w_k p_{ik} \right)^2 + \lambda_w \sum_k \left(w_k - \frac{1}{K} \right)^2 \right\}, \quad \lambda_w = 0.01. \quad (88)$$

The objective is Brier loss. Nonnegative simplex constraints make the ensemble a convex combination and limit unstable extrapolation.

14.6 Purging and Embargo

Let observation i have feature date t_i and target end u_i . For a later stage beginning at T , purge requires $u_i < T$. With embargo E , require $t_i < T - \text{BDay}(E)$ where applicable; the reference setting is $E = 252$ business days. FinCompass uses such boundaries between component calibration, stacking, final calibration and locked evaluation rather than assuming non-overlapping row indices are sufficient.

14.7 Moving Date-Block Bootstrap

Let $D_1, \dots, D_m$ be unique observation dates and let B be block length. A bootstrap replicate samples starting indices and concatenates contiguous date blocks, carrying all cross-sectional rows attached to each selected date. This preserves same-date dependence and approximates serial dependence from overlapping targets. For $H = 252$ and monthly-like sampling step $s = 21$, the automatic reference block size is $\lceil 252/21 \rceil = 12$ dates.

14.8 Adaptive Residual

Let p_t^A be frozen anchor probability, $z_t \in \mathbb{R}^{13}$ the realtime feature vector, and m_t posterior coefficient mean. Then

$$\Delta_t^{\text{raw}} = z_t^\top m_t, \quad (89)$$

$$\Delta_t = \text{clip}(\Delta_t^{\text{raw}}, -\delta_{\max}, \delta_{\max}), \quad (90)$$

$$p_t^C = \text{clip}_p \left[\text{sigmoid}\{\text{logit}(p_t^A) + \Delta_t\} \right]. \quad (91)$$

Posterior shift variance is $z_t^\top P_t z_t$.

14.9 Sequential Covariance Update

After covariance discount/process noise,

$$P^- = P/\lambda + qI. \quad (92)$$

A local logistic Hessian contribution is $wzz^\top$ with theoretical curvature $p(1-p)$. The executable update uses

$$w = \max\{p(1-p), 10^{-6}\} \quad (93)$$

to avoid numerical degeneracy near probability boundaries. Direct inversion of $((P^-)^{-1} + wzz^\top)$ for every sample is unnecessary. The Sherman–Morrison–Woodbury identity gives

$$\left((P^-)^{-1} + wzz^\top \right)^{-1} = P^- - \frac{wP^-zz^\top P^-}{1 + wz^\top P^-z}. \quad (94)$$

Set $v = P^-z$ and $d = 1 + wz^\top v$ to obtain the implemented update

$$P' = P^- - \frac{w}{d} vv^\top. \quad (95)$$

The corresponding one-step mean update is

$$m' = m + \frac{v}{d}(y - p). \quad (96)$$

The implementation symmetrizes P' numerically and applies a small eigenvalue floor if required. This is a local Gaussian/Laplace (quasi-Newton) approximation for a non-conjugate logistic likelihood, not an exact Bayesian posterior. If deployment clipping is active, the update remains tied to the local logistic approximation used by the state update rather than to the derivative of the clipped output map.

14.10 Date-Balanced Proper Scoring

Let $\mathcal{D}$ be unique observation dates and n_d the number of matured rows on date d . Give each row weight

$$\omega_i = \frac{1}{|\mathcal{D}|n_{d(i)}}. \quad (97)$$

Then date-balanced Brier score is

$$B = \sum_i \omega_i (p_i - y_i)^2. \quad (98)$$

Date-balanced log loss is

$$L = - \sum_i \omega_i [y_i \log p_i + (1 - y_i) \log(1 - p_i)]. \quad (99)$$

ECE uses the same weights inside probability bins.

14.11 Fail-Closed Application

Let $A_t \in \{0, 1\}$ indicate live eligibility of the frozen anchor and let $G_t \in \{0, 1\}$ combine the statistical adaptive gate, adaptive-application enablement, and recent market-provider verification. Then

$$p_t^{\text{live}} = \begin{cases} \text{unavailable}, & A_t = 0, \\ p_t^C, & A_t = 1 \text{ and } G_t = 1, \\ p_t^A, & A_t = 1 \text{ and } G_t = 0. \end{cases} \quad (100)$$

SEC and macro source staleness is handled at the feature layer by zeroing source-dependent inputs; it is not conflated with anchor eligibility. The equation therefore matches the application's fail-closed control flow.

14.12 Wyckoff Hidden-State Extension

If future work defines phase state $C_t \in \{A, M, D, R\}$ and obtains defensible point-in-time labels or a generative model, a hidden Markov filter could use transition matrix T and emission likelihoods $L_k(x_t)$:

$$\pi_t(k) = \frac{L_k(x_t) \sum_j T_{jk} \pi_{t-1}(j)}{\sum_\ell L_\ell(x_t) \sum_j T_{j\ell} \pi_{t-1}(j)}. \quad (101)$$

The denominator normalizes probabilities to one. The challenge is empirical identification and validation, not algebra. The current application therefore does not make this claim.

15 Implementation and Reproducibility

This section connects the manual to the reviewed FinCompass source tree. File names are included so a technical reader can trace a displayed quantity back to executable code.

15.1 Quick Local Start

From the application directory:

```
python -m venv .venv
# Activate the environment for your operating system
pip install -r requirements.txt
uvicorn api:app --host 127.0.0.1 --port 8000
```

Then open <http://127.0.0.1:8000/> in a browser.

The release verifier can be run with development requirements installed:

```
pip install -r requirements-dev.txt
python tools/verify_release.py
```

15.2 Code-to-Equation Map

Manual concept	Source	Purpose
Continuous metric scoring, Beta evidence, dependence envelope	<code>services/scoring.py</code>	Evidence pillars, coverage, 90% model interval, confidence.
Evidence constants and pillar weights	<code>config.py</code>	Prior (2, 2), evidence scale 4, 6000 draws, 90% level, 25/20/20/20/15 weights.
Backward-looking anchor features	<code>forecasting/features.py</code>	Return, relative return, volatility, drawdown, trend and volume features; target construction.
Point-in-time SEC fundamentals	<code>forecasting/sec_fundamentals.py</code>	Filing-availability semantics for historical features.
Bayesian logistic model	<code>forecasting/bayesian.py</code>	MAP optimization, Gaussian prior, Hessian covariance, posterior intervals.
Calibration	<code>forecasting/calibration.py</code>	Sigmoid/isotonic probability calibration.
Frozen ensemble and validation gate	<code>forecasting/model.py</code>	Three validation stages, stacking, locked test, bootstrap, walk-forward evaluation.
Forecast settings	<code>forecasting/config.py</code> and <code>config/*.json</code>	Horizon, benchmark, hurdle, priors, calibration and validation thresholds.
Realtime 13-feature vector	<code>realtime/adaptive.py</code> , <code>realtime/features.py</code>	Candidate shift and explainable feature contributions.
Realtime event providers	<code>realtime/providers.py</code>	Market, filing and macro event acquisition/timestamps.
Realtime state storage	<code>realtime/store.py</code>	Pending labels, source checks, state persistence.
Adaptive settings	<code>realtime/config.py</code> , <code>config/realtime-*.json</code>	Shift bound, half-life, forgetting, process noise, gate thresholds.
Exact-horizon resolution / service orchestration	<code>services/realtime_service.py</code>	Applies target contract, updates matured labels, returns live snapshot.
API	<code>api.py</code>	Versioned evidence/forecast/realtime/settings endpoints.
Release regression checks	<code>tests/</code> , <code>tools/verify_release.py</code>	Guards scoring, temporal semantics, security, frontend and artifact integrity.

15.3 Configuration Lineage

A configuration file is part of a model's meaning. If target-defining settings change, the old model cannot be silently reused. A model manifest therefore records the relevant settings, data provenance, split metadata and artifact hash.

The adaptive layer goes further. Its state key is constructed from the base-model identity, realtime engine

version and adaptive-settings fingerprint. In schematic form,

$$K = \text{SHA256}(\text{base model} \parallel \text{engine} \parallel \text{settings})_{1:24}. \quad (102)$$

This prevents a posterior learned under one forgetting factor or shift bound from being interpreted as if it belonged to another configuration.

15.4 Source Timestamps

A realtime event has at least three different times:

1. when the underlying market/filing/macro event occurred;
2. when FinCompass received or constructed the event;
3. when the provider was last checked successfully.

These times answer different questions. Event age controls decay. Provider-check age controls whether the system can safely conclude that no newer event has been missed.

15.5 Data Provenance and Model Identity

The release includes SHA-256 manifests for datasets/models. The purpose is not cryptographic prediction; it is reproducibility. If a file changes, the hash changes. A reviewer can therefore detect whether the model or dataset used to produce a report is the same artifact that was originally validated.

15.6 Wyckoff Implementation Boundary

The Wyckoff chapters in this manual are not currently a source-code module. They are an educational interpretation layer. If later implemented, a new module should have:

- an explicit feature contract;
- point-in-time construction tests;
- a declared phase-labeling protocol;
- separate train/validation/test partitions;
- calibration tests if probabilities are reported;
- its own model version and artifact hash;
- no automatic authority over the validated anchor until activation criteria are met.

16 Glossary

Term	Plain-language meaning
Accumulation	A trading-range hypothesis in which selling pressure is being absorbed before a possible upward trend. Not every range is accumulation.
Adaptive residual	A bounded realtime adjustment added to the frozen anchor's log-odds when new information arrives.
Anchor probability	The frozen calibrated probability from the validated historical forecasting model.

Term	Plain-language meaning
Automatic rally (AR)	A rebound after a selling climax that helps establish the upper part of an accumulation range.
Automatic reaction (AR)	In distribution language, the decline after a buying climax that helps establish the lower range boundary.
Bayesian prior	A probability distribution representing the model's state before the current evidence is incorporated.
Brier score	Mean squared error of probability forecasts; lower is better.
Calibration	Agreement between predicted probabilities and observed event frequencies.
Candidate probability	Anchor probability after the realtime residual is applied, before the activation gate decides whether it may be used live.
Comonotonic	Perfect positive rank dependence; used in the evidence layer as a conservative sensitivity scenario.
Credible interval	Bayesian interval containing a chosen amount of posterior probability under the model.
DCA	Dollar-cost averaging: investing planned amounts at regular intervals. It does not guarantee profit or prevent loss.
Distribution	A trading-range hypothesis in which supply increasingly offsets demand near a mature advance.
Drawdown	Percentage decline from a prior peak.
ECE	Expected calibration error; a summary of the gap between predicted probabilities and observed frequencies.
Embargo	A deliberate time gap around validation/test boundaries to reduce leakage from overlapping outcomes.
Evidence coverage	Fraction of expected metric weight actually observed in the evidence score.
Evidence score	FinCompass's 0–10 Bayesian research summary; it is not a return probability.
Excess return	Security return minus benchmark return over the same endpoints.
Filtration $\mathcal{F}_t$	Mathematical notation for all information available by time t .
Fixture-only	Validation tier indicating synthetic/regression validation of machinery, not empirical market forecasting authority.
Forgetting factor	Online-learning parameter that gradually discounts old posterior information.
Horizon	Number of future common trading sessions used to define the target.
LPS	Last point of support, a pullback/retest in an accumulation/markup interpretation.
LPSY	Last point of supply, a weak rally/retest in a distribution interpretation.
Log loss	Proper probability score that penalizes confident wrong forecasts strongly.
Logit	Natural logarithm of probability odds, $\ln[p/(1 - p)]$.
Markup	A sustained upward market phase following a successful accumulation/re-accumulation hypothesis.
Markdown	A sustained downward market phase after supply dominates.
MAP	Maximum a posteriori estimate: parameter values maximizing the posterior density.
Point in time	Data rule requiring that each historical feature was actually available at the historical observation date.

Term	Plain-language meaning
Posterior	Updated probability distribution after evidence/data are incorporated.
Prequential	“Predict, then evaluate when the outcome arrives.” Online model quality is judged sequentially without learning from the future label first.
Probability clip	Lower/upper bound preventing exactly 0 or 1 probabilities.
Process noise	Small covariance inflation allowing the adaptive model to remain responsive to changing relations.
Purge	Removal of observations whose outcome windows overlap a later validation/test segment.
Relative strength	Security performance compared with a benchmark, not absolute price movement alone.
SC	Selling climax.
Secondary test (ST)	A revisit of a range boundary after a climax/reaction to assess whether prior pressure remains.
Sigmoid	Function mapping any real number to (0, 1); inverse of logit.
Sign of strength (SOS)	Strong upward behavior out of/within an accumulation structure.
Sign of weakness (SOW)	Strong downward behavior in a distribution structure.
Spring	Optional accumulation event in which price breaks support and recovers; can fail.
Stacking	Learning nonnegative weights for multiple calibrated model probabilities.
UT / UTAD	Upthrust / upthrust after distribution: an upside break that fails back into a range, interpreted only in context.
Validation tier	Governance state controlling what claims/use a model is allowed to make.
Wyckoff	A price-volume market-structure framework associated with Richard D. Wyckoff; used here descriptively unless separately validated.

A Formula Sheet

This appendix collects the principal equations without the surrounding explanation.

Returns and relative behavior

$$\begin{aligned}
 R &= P_1/P_0 - 1, \\
 R^{\text{excess}} &= R_{\text{stock}} - R_{\text{benchmark}}, \\
 r_{\text{range}} &= (C - L)/(H - L), \\
 RV &= V/\tilde{V}, \\
 RS &= (H_t - L_t)/\tilde{S}.
 \end{aligned}$$

Probability

$$\begin{aligned}
 \text{odds}(p) &= p/(1 - p), \\
 \text{logit}(p) &= \ln[p/(1 - p)], \\
 \text{sigmoid}(x) &= 1/(1 + e^{-x}).
 \end{aligned}$$

Evidence scoring

$$\begin{aligned}
 f(x) &= s_k + \frac{x - a_k}{a_{k+1} - a_k} (s_{k+1} - s_k), \\
 z^{IQR} &= d \frac{x - m}{\max(IQR, \epsilon)}, \\
 s^{peer} &= \text{clip}(5 + 2.2z^{IQR}, 1, 9.5), \\
 s &= (1 - \rho)s^{abs} + \rho s^{peer}, \\
 \alpha &= \alpha_0 + c \sum_j w_j q_j, \\
 \beta &= \beta_0 + c \sum_j w_j (1 - q_j), \\
 S &= 10\alpha / (\alpha + \beta), \\
 C &= \min(1, W^{obs} / W^{exp}).
 \end{aligned}$$

Forecast target

$$\begin{aligned}
 R_{i,t}^{(H)} &= P_i(\tau_H(t)) / P_i(t) - 1, \\
 Y_{i,t} &= \mathbb{I}\{R_{i,t}^{(H)} - R_{b,t}^{(H)} > \eta\}, \\
 p_{i,t} &= \mathbb{P}(Y_{i,t} = 1 \mid \mathcal{F}_t).
 \end{aligned}$$

Bayesian logistic model

$$\begin{aligned}
 p_i &= \text{sigmoid}(x_i^\top \beta), \\
 \mathcal{L}(\beta) &= \sum_i [\log(1 + e^{x_i^\top \beta}) - y_i x_i^\top \beta] + \frac{1}{2} \beta^\top \Lambda \beta, \\
 H &= X^\top W X + \Lambda, \\
 \beta \mid D &\approx N(\hat{\beta}, H^{-1}).
 \end{aligned}$$

Calibration and ensemble

$$\begin{aligned}
 \hat{p} &= \text{sigmoid}\{a \logit(p^{raw}) + b\}, \quad p^{raw} \in [10^{-6}, 1 - 10^{-6}]. \\
 \min_{w \in \Delta_K} &\left\{ \frac{1}{n} \sum_i \left(y_i - \sum_k w_k p_{ik} \right)^2 + 0.01 \sum_k \left(w_k - \frac{1}{K} \right)^2 \right\}.
 \end{aligned}$$

Realtime residual

$$\begin{aligned}
 \Delta^{raw} &= z^\top m, \\
 \Delta &= \text{clip}(\Delta^{raw}, -\delta_{max}, \delta_{max}), \\
 p^C &= \text{clip}_p\{\text{sigmoid}[\logit(p^A) + \Delta]\}, \\
 \text{Var}(\Delta) &= z^\top P z, \\
 f(a) &= 2^{-a/h_{1/2}}.
 \end{aligned}$$

Sequential update

$$\begin{aligned}
 P^- &= P/\lambda + qI, \\
 w &= \max\{p(1-p), 10^{-6}\}, \\
 v &= P^- z, \\
 d &= 1 + wz^\top v, \\
 P' &= P^- - (w/d)vv^\top, \\
 m' &= m + (v/d)(y-p).
 \end{aligned}$$

Probability validation

$$\begin{aligned}
 \text{Brier} &= n^{-1} \sum_i (p_i - y_i)^2, \\
 \text{LogLoss} &= -n^{-1} \sum_i [y_i \ln p_i + (1 - y_i) \ln(1 - p_i)], \\
 \text{ECE} &= \sum_b \omega_b |\bar{p}_b - \bar{y}_b|.
 \end{aligned}$$

Fail-closed live output

$$p^{\text{live}} = \begin{cases} \text{unavailable}, & A = 0, \\ p^C, & A = 1, G = 1, \\ p^A, & A = 1, G = 0. \end{cases}$$

Wyckoff descriptive measurements

$$\begin{aligned}
 r_t &= (C_t - L)/(H - L), \\
 I_{\text{reject}} &= \mathbb{I}\{L_t < L\} \mathbb{I}\{C_t > L\}, \\
 I_{\text{upthrust}} &= \mathbb{I}\{H_t > H\} \mathbb{I}\{C_t < H\}, \\
 b_t &= (C_t - H)/H, \\
 q_t &= (L_t^{\text{pull}} - H)/H.
 \end{aligned}$$

B Student Exercises and Worked Solutions

The exercises are ordered from introductory arithmetic to graduate-level reasoning.

Exercise 1: return

A stock rises from \$40 to \$46. What is the simple return?

Solution.

$$R = 46/40 - 1 = 1.15 - 1 = 0.15 = 15\%.$$

Exercise 2: benchmark outperformance

A stock returns 9%; the benchmark returns 6%. With hurdle 2%, what is the target label?

Solution. Excess return is $9 - 6 = 3\%$. Since $3\% > 2\%$, $Y = 1$.

Exercise 3: range position

A trading range is \$25–\$35 and current close is \$28. Find r .

Solution.

$$r = (28 - 25)/(35 - 25) = 3/10 = 0.30.$$

Price is 30% of the way up from support.

Exercise 4: relative volume

Median recent volume is 1.5 million; current volume is 2.4 million. Find RV .

Solution.

$$RV = 2.4/1.5 = 1.6.$$

Current volume is 60% above the median reference.

Exercise 5: interpolation

A metric has knots (0.08, 6.4) and (0.12, 8.0). Find score at 0.11.

Solution.

$$\begin{aligned} f(0.11) &= 6.4 + \frac{0.11 - 0.08}{0.04}(1.6) \\ &= 6.4 + 0.75(1.6) = 7.6. \end{aligned}$$

Exercise 6: Beta evidence update

Start with Beta(2, 2). One metric has score 8/10, evidence weight 1, and scale $c = 4$. Find updated (α, β) and posterior 0–10 score.

Solution. $q = 0.8$, effective weight 4.

$$\alpha = 2 + 4(0.8) = 5.2, \quad \beta = 2 + 4(0.2) = 2.8.$$

$$S = 10(5.2/8) = 6.5.$$

Although the metric score is 8, the posterior is only 6.5 because one piece of evidence is shrunk toward the neutral prior.

Exercise 7: log-odds update

Anchor is 0.60 and residual shift is -0.25 . Find candidate probability.

Solution.

$$\text{logit}(0.60) = \ln(1.5) \approx 0.4055.$$

New log-odds = $0.4055 - 0.25 = 0.1555$.

$$p^C = \text{sigmoid}(0.1555) \approx 0.5388.$$

Candidate is about 53.9%.

Exercise 8: residual clipping

Anchor is 0.50, raw shift is 1.20, and $\delta_{max} = 0.75$. Find bounded shift and candidate.

Solution. Bounded shift is 0.75. Since $\text{logit}(0.5) = 0$,

$$p^C = \text{sigmoid}(0.75) \approx 0.6792.$$

Exercise 9: Brier score

Forecasts (0.7, 0.4, 0.9, 0.2) have outcomes (1, 0, 0, 0). Find Brier score.

Solution. Squared errors are 0.09, 0.16, 0.81, 0.04. Sum = 1.10.

$$\text{Brier} = 1.10/4 = 0.275.$$

The confident wrong 0.9 forecast contributes most of the loss.

Exercise 10: exact endpoint reasoning

A 252-session forecast matures on 15 June, but the label resolver runs on 30 June. Which date's price belongs to the label?

Solution. The exact 252nd common trading session, 15 June. The scheduler's later run date does not redefine the event.

Exercise 11: date-balanced weights

Day A has 50 observations and Day B has 10. There are two dates. What weight does each row receive?

Solution. Day A row weight is

$$1/(2 \cdot 50) = 0.01.$$

Day B row weight is

$$1/(2 \cdot 10) = 0.05.$$

Each day therefore has total weight 0.5.

Exercise 12: support rejection

Support is 80, daily low 77.6, close 81.5. Calculate break depth and rejection indicator.

Solution.

$$d = (80 - 77.6)/80 = 0.03 = 3\%.$$

Because $\text{low} < 80$ and $\text{close} > 80$, the simple rejection indicator equals 1. This is a spring-like candidate, not confirmation.

Exercise 13: portfolio trim

Portfolio is \$150,000. A position is 16% and the declared cap is 10%. Ignoring taxes/costs, how much value would be trimmed to rebalance exactly to the cap if total portfolio value is treated as fixed for the exercise?

Solution. Current value = $0.16(150,000) = 24,000$. Target = 15,000. Difference = 9,000.

Exercise 14: hidden-state reasoning

Why can Equation (64) not be called a validated Wyckoff probability merely because it sums to one?

Solution. Mathematical normalization creates a probability distribution *conditional on the assumed model*. It does not establish that phase labels are well defined, that emission likelihoods are correct, that transitions match real markets, or that the resulting probabilities are calibrated out of sample. Empirical validation is still required.

Exercise 15: thesis versus price

A stock falls 50% while earnings fall 60%. Old price/earnings ratio was 20. Is the stock automatically cheaper on earnings?

Solution. Let old price be 100 and old earnings be 5, giving P/E 20. New price is 50; new earnings are 2. New P/E is 25. The price is lower, but the valuation multiple is higher. “Down 50%” is not sufficient evidence of cheapness.

C Operational checklists

Before trusting an evidence score

1. Confirm ticker/company and data source.
2. Check fetched/as-of time.
3. Review individual pillars, not only composite.
4. Check evidence coverage.
5. Check 90% model interval and confidence label.
6. Look for missing/abnormal metrics.
7. Remember that evidence-score probabilities concern the score, not future return.

Before using a forecast probability

1. Verify model tier is live-eligible.
2. Verify benchmark, horizon and hurdle match the question.
3. Verify as-of time and point-in-time data contract.
4. Review calibration/proper-score evidence and uncertainty.
5. Confirm the model is not a synthetic fixture.
6. If Live differs from Forecast, inspect gate status and source freshness.

Before staging accumulation into a single company

1. Is the business thesis intact?
2. Is evidence coverage sufficient for the conclusion?
3. Is balance-sheet risk acceptable?
4. Is the position/sector below declared concentration caps?
5. Is the security still in obvious markdown, or is there reproducible evidence of stabilization?
6. If using Wyckoff language, identify range boundaries before labeling a spring/SOS/LPS.
7. If a live-eligible forecast exists, is it supportive or contradictory?
8. Define maximum total capital and tranche sizes before entering.
9. Define what evidence would stop further tranches.
10. Do not borrow or use emergency cash merely because a score is high.

Before deciding to hold

1. Has the thesis changed?
2. Is position size still acceptable?
3. Has valuation become extreme relative to the plan?
4. Is the trend/market structure still healthy?
5. Has the validated forecast materially deteriorated?
6. Would selling create taxes/costs without reducing a meaningful risk?

Before trimming

1. Is the reason portfolio concentration, valuation, thesis deterioration, distribution-like structure, liquidity need, or a combination?
2. Is the trim amount tied to a declared target weight rather than emotion?
3. Is the decision all-or-nothing when a partial rebalance would address the risk?
4. Are taxes and transaction costs considered outside FinCompass?
5. If market structure is the main reason, has the pattern persisted beyond one failed breakout?

Before calling a Wyckoff phase

1. Mark support and resistance before interpreting the latest bar.
2. Measure range position.
3. Compare volume with a declared reference window.
4. Compare spread/result with normal behavior.
5. Examine relative strength versus benchmark.
6. Require follow-through or a test where appropriate.

7. Record alternative hypothesis: re-accumulation, redistribution, continuation or noise.
8. Do not convert the phase label into a numeric return probability without separate validation.

D Validation Evidence and Sources

What the bundled fixtures establish

FinCompass ships deterministic historical-style and adaptive streaming fixtures so regression tests can verify the statistical pipeline, temporal rules, calibration metrics, state lineage, source freshness behavior and fail-closed gates. These fixtures are deliberately classified `fixture_only`. Their successful metrics are evidence that the software behaves as designed on controlled data; they are not evidence that a real portfolio will outperform.

What a real market validation set must add

A market-grade study needs, at minimum:

- point-in-time company fundamentals based on actual public availability;
- corporate-action-adjusted historical prices;
- historical universe membership to control survivorship bias;
- delisted securities and delisting outcomes;
- benchmark data on the same endpoint contract;
- vintage macro data where revisions matter;
- predeclared feature/target definitions;
- frozen development, calibration and locked-test protocol;
- a record of every model variant tried so research multiplicity is visible.

Wyckoff source note

Wyckoff's original writings are historical market literature, not modern statistical validation studies. Wyckoff [1910] is a primary historical source for tape-reading ideas. Modern tutorials such as StockCharts and Wyckoff Analytics organize the now-familiar accumulation/distribution phase language [StockCharts ChartSchool, 2026, Wyckoff Analytics, 2026]. The manual cites them for definitions and historical method, not as proof of predictive efficacy.

DCA source note

Investor.gov defines dollar-cost averaging as investing equal portions at regular intervals through market fluctuations [U.S. Securities and Exchange Commission, Investor.gov, 2026]. The Vanguard research cited in this manual addresses a different question: whether a lump sum available now should be invested immediately or temporarily staged. Its historical result favors immediate investment more often because cash has an opportunity cost, while acknowledging behavioral/risk reasons some investors may prefer staging [Wimmer and Zorn, 2023].

Probability and validation reading

For Bayesian modeling, Gelman et al. [2013] provides a broad reference. The Brier score originates with Brier [1950]. The prequential idea of judging predictions sequentially is associated with Dawid [1984]. Financial model-selection risk and data snooping are discussed by White [2000], Harvey et al. [2016] and Bailey et al. [2017].

Official data sources

SEC EDGAR APIs provide public filing data and are the relevant official source for filing submissions used in point-in-time company-data workflows [U.S. Securities and Exchange Commission, 2026]. ALFRED/FRED provides vintage-aware macroeconomic series useful for avoiding revision hindsight [Federal Reserve Bank of St. Louis, 2026].

Final reading rule

When sources disagree, prefer the source closest to the fact being asserted: official filing for company disclosures, official macro vintage for historical macro data, reviewed implementation artifacts for FinCompass behavior, and independent statistical literature for validation methodology. A chart label should never outrank a broken data contract.

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